

Schooled by Trade?

Retraining and Import Competition *

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Abstract

International trade creates winners and losers, yet surprisingly little is known about policies that help workers escape declining sectors. We study how government-subsidized retraining in Germany can help workers smooth sectoral shocks. Using administrative data, we show that workers switch sectors and retrain in response to Chinese import competition. We introduce retraining into a dynamic model of labor mobility within a quantitative trade framework, where workers simultaneously choose their sector and whether to retrain; we find retraining lowers the costs of moving between sectors. In counterfactual experiments, retraining compresses the distributional effects of trade shocks while leaving aggregate gains unchanged.

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Introduction

International trade creates winners and losers, and whether an individual worker wins or loses is closely tied to their sector of employment and its exposure to import competition.¹ Yet, despite enormous interest from policymakers, surprisingly little is known about policies that help workers escape declining sectors and thus blunt trade’s uneven effects. With new sectoral shocks — trade wars, supply chain disruptions, the rollout of artificial intelligence, and climate change ([Dingel and Meng 2025](#)) — on the horizon, the need for such policies has rarely been greater.

We study how government-subsidized retraining in Germany can help workers adjust to trade shocks. Using administrative data, we show that workers switch sectors and retrain in response to Chinese import competition. We introduce retraining into a quantitative general equilibrium model of labor supply and trade, where workers simultaneously choose their sector and whether to retrain; our calibration reveals that retraining lowers the costs of moving between sectors. In counterfactual experiments, lower-cost retraining speeds up short-run adjustment to the “China shock” while dampening long-run sectoral shifts and compressing the distributional effects of trade. Retraining thus offers policymakers a lever that accelerates adjustment to trade shocks while mitigating their uneven impacts.

We begin by documenting salient features of Germany’s extensive worker retraining system. While Germany’s Employment Agency provides vouchers to cover participants’ costs, private organizations typically provide the courses themselves, which deliver a mix of sector-specific skills and job-finding assistance. In the Sample of Integrated Labor Market Biographies (SIAB), we find that retraining is common but brief: roughly a quarter of workers retrain over the course of their careers, and the median retraining spell lasts just under six weeks. Retraining is tightly linked to transitions across sectors: even after controlling for a broad set of worker characteristics, workers who retrain are more than twice as likely to switch sectors over the following decade than those who do not.

We next show how workers use retraining to adjust to trade shocks, in particular rising import competition from China. Manufacturing workers exposed to this “China shock” suffered relative declines in employment and earnings—and responded by switching sectors and, crucially, retraining at higher rates. Together with the strong correlation between retraining and sector switching, this pattern suggests that workers find retraining useful in making transitions across sectors. Yet the simultaneous nature of the decisions to retrain and switch sectors makes it hard to identify exactly which way causality runs from the data alone.

We therefore introduce a new theory of retraining and sector switching that addresses this identification challenge and allows us to quantify the aggregate gains from retraining in general equilibrium. In particular, we build on the [Gentzkow \(2007\)](#) model of complementary choices and embed it within the dynamic sectoral choice framework of [Artuç, Chaudhuri, and McLaren \(2010\)](#). The central innovation is that workers simultaneously choose their sector and whether to retrain;

¹See e.g. [Artuç, Chaudhuri, and McLaren \(2010\)](#), [Dix-Carneiro \(2014\)](#), [Fort, Pierce, and Schott \(2018\)](#), [Traiberman \(2019\)](#), and [Adão et al. \(2022\)](#).

workers face deterministic switching costs, and the payoff to retraining depends on whether a worker enters a new sector. We show that, in the natural case when retraining *complements* sector switching, expanding access to retraining acts like a reduction in sector-switching costs and raises worker mobility across sectors. In the wake of trade shocks, workers in adversely-affected sectors thereby enjoy greater option values of later switching – at lower cost – to growing sectors. As a result, retraining compresses the distribution of welfare changes across workers but, because these shifts net out, leaves the aggregate gains from trade unchanged. This compression of the welfare distribution weakens workers’ incentives to reallocate across sectors and means that retraining ultimately shrinks the long-run sectoral employment response to trade shocks, but we also show that the speed of the transition following such a shock rises with the retraining rate.

To quantify these insights, we enrich our model with worker heterogeneity by age and skill and embed it within the quantitative trade framework of [Caliendo and Parro \(2015\)](#). We calibrate retraining costs, the dispersion of idiosyncratic sector switching and retraining preferences, and the degree of complementarity – or substitutability – between retraining and switching to match key features of the data in an iterative procedure. In particular, we match the model’s predicted retraining rates, responses to the China shock, and correlations between retraining and switching to their empirical counterparts. We find – as expected – that retraining and sector-switching are complements. A worker who retrains faces switching costs that are on average 30% lower than one who does not, and as a result is roughly twice as likely to move to a new sector.

Finally, we use our calibrated model to quantify how lower-cost retraining alters the aggregate and distributional effects of trade shocks, including the China shock of the 2000s as well as a hypothetical “Trump tariff” on German exports to the United States. Focusing on the China shock, going from an economy with no retraining to one with retraining costs low enough to triple the status-quo retraining rate compresses the distribution of the shock’s welfare effects by 23%. The worst-hit workers’ losses shrink by 26% — but, in line with our analytical predictions, retraining scarcely changes the China shock’s aggregate effects. Retraining has more muted effects on the speed of the employment transition, in part because general equilibrium changes in real wages strongly offset the direct effect of lower switching costs on labor mobility. In contrast, we find that US tariffs on German exports have fairly uniform negative effects across manufacturing sectors, leaving little scope for retraining to blunt workers’ exposure.

Overall, we find that retraining is an attractive tool for policymakers concerned about the distributional consequences of international trade. Germany’s system of government-subsidized retraining limits adversely-affected workers’ losses and speeds up labor market adjustment, all without diminishing the aggregate gains from trade. At the same time, the magnitudes implied by our quantitative model are moderate, and the effectiveness of retraining depends on whether a particular shock creates new opportunities in growing sectors – as with the China shock – or is uniformly negative – as with the Trump tariffs. Retraining therefore complements rather than substitutes for other adjustment policies.

Related Literature

A now-voluminous literature emphasizes the distributional conflicts created by international trade (Autor, Dorn, and Hanson 2013; Dauth, Findeisen, and Suedekum 2014), whether they play out across sectors (Dauth, Findeisen, and Suedekum 2021), skill levels (Antràs, Gortari, and Itskhoki 2017; Lee 2020; Traiberman 2019), space (Galle, Rodriguez-Clare, and Yi 2021; Adao, Arkolakis, and Esposito 2026), or even firms (Schott, Pierce, and Tello-Trillo 2020). Moreover, substantial barriers impede worker mobility across sectors or space (Dix-Carneiro 2014; Caliendo, Dvorkin, and Parro 2019). We provide some of the first evidence on the gains from a policy that moderates these unequal effects of trade.

Simultaneously, extensive empirical evidence quantifies the micro-level effects of worker training programs (McCall, Smith, and Wunsch 2016). While randomized experiments in developing countries find large causal effects on re-employment and firm switching (Alfonsi et al. 2020; Alfonsi et al. 2024), studies in the German context generally find mixed effects (Lechner, Miquel, and Wunsch 2011; Osikominu 2013; Biewen et al. 2014; Fitzenberger, Osikominu, and Paul 2023). More recently, Humlum, Munch, and Rasmussen (2025) use a judge-IV design to show that training programs in Denmark are effective in moving workers into employment, particularly in new occupations. Our structural model allows us to move beyond worker-level effects to quantify aggregate and distributional effects as well as interactions with sectoral shocks.

Finally, a nascent literature studies education as a margin of adjustment to labor market shocks. Dix-Carneiro (2014) studies a hypothetical retraining policy within a quantitative model of trade and labor market dynamics. Recent empirical contributions estimate the worker-level effects of Trade Adjustment Assistance (TAA) subsidies in the US (Hyman 2018), and Schulze (2024) models community college enrollment during the Great Recession. Smeets, Tian, and Traiberman (2025) and Adachi and Skipper (2025) both embed educational choices — by college students entering the labor market for the first time, and by adults entering long vocational training programs, respectively — in dynamic discrete choice models and explore how they shape the effects of labor market shocks. In contrast, we contribute a general equilibrium framework centered around the joint decision of existing workers to retrain and enter a new sector. As a result, we can quantify how retraining alters the welfare effects of trade or other sectoral shocks, taking into account changes in prices and wages.

Outline

The remainder of the paper proceeds as follows. Section 1 describes the data and the institutional context of retraining in Germany. Section 2 presents our main empirical results on the interaction of import competition and retraining. Section 3 develops a theory of retraining, labor mobility, and trade. Section 4 calibrates the model’s key parameters, and Section 5 uses the model to perform counterfactual experiments. Finally, Section 6 concludes.

1 Retraining in Germany

German workers benefit from comprehensive government-sponsored retraining programs. In this section, we introduce the administrative labor market data with which we measure retraining and then discuss the institutional details of retraining in Germany.

1.1 Data

We observe workers' labor market outcomes and characteristics in the weakly anonymous version of the German Sample of Integrated Labour Market Biographies (SIAB) 1975-2021 ([Graf et al. 2023](#)).² The SIAB draws a 2% sample from Germany's primary administrative employment records, known as the Integrated Employment Biographies (IEB). The resulting worker-level panel records, for each worker and month between 1975 and 2021, all active employment spells, unemployment benefits, and participation in "active labor market policies" such as retraining ([Schmucker, Seth, and Berge 2023](#)).

To compute workers' trade exposure, we combine this labor market data with trade flows from the UN's Comtrade Database.³ Following [Autor, Dorn, and Hanson \(2013\)](#) and [Dauth, Findeisen, and Suedekum \(2021\)](#), we focus on trade flows between China and Germany and, for identification, additionally draw on trade flows between a number of other high-income countries and China. Supplemental Appendices [A.1.1-A.1.2](#) provide additional details regarding each data source as well as our cleaning and preparation procedures.

1.2 Institutional Context

The German Federal Employment Agency (BA) manages a variety of active labor market policies to support workers' employment prospects. These policies target workers across all career stages and regardless of their current employment status ([Dauth 2020](#); [Fitzenberger, Osikominu, and Paul 2023](#)).

We focus on three categories of programs which we collectively define as *retraining* and which comprise two-thirds of active labor market policy episodes during our sample period. First, *activation and vocational integration* measures match workers with jobs and provide necessary skills, from job application training to foreign language classes, to re-enter the labor market. Second, *career choice and vocational training* programs offer career advising and tutoring as well as government-sponsored internships ([Bundesagentur für Arbeit 2021a](#); [Roesler et al. 2021](#)). Third, *vocational retraining and further education* programs range from short job search or computer skills courses to year-long training in more general occupational skills, e.g. marketing ([Osikominu 2013](#); [Biewen et al. 2014](#)).⁴

²Data access was provided via on-site use at the Research Data Centre (FDZ) of the German Federal Employment Agency (BA) at the Institute for Employment Research (IAB) and subsequently via remote data access.

³<https://comtradeplus.un.org>.

⁴These three program types correspond to the *erwstat* codes 10001 (activation and vocational integration), 10002 (career choice and vocational training), and 10003 (vocational retraining and further education) in the Participants-In-

Table 1: Retraining Curriculum Examples

Topic	<i>High-voltage fundamentals for electric vehicles</i>	<i>Career change: vehicle technology</i>
Provider	TÜV Rheinland Academy	TÜV Rheinland Academy
Length	1–4 weeks	3–6 months
Prerequisites	Interest in electric mobility	Interest in vehicle technology, ideally with initial experience
Degree Obtained	Completion certificate	Completion certificate
Cost with Subsidy	0	0
Contents	• Lithium-ion battery systems • Power converter applications • Hybrid and electric drivetrains • Charging technologies • Electrical hazards and safety • Relevant laws and regulations • Practical component	• Local job, training opportunities • Vehicle maintenance and repair • Safety inspections • Defect identification • Custom features and accessories • Driver assistance systems • Connected vehicle systems • Applied mathematics • Technical German vocabulary

Notes: Table reproduces summary curricula for two courses for which workers could use government-provided retraining vouchers. Both example courses are drawn from among the offerings of [TÜV Rheinland Akademie GmbH](#), a regional private provider of worker training courses. Sources: [Grundlagen Hochvolttechnik](#) and [Berufseinstieg Fahrzeugtechnik](#).

After a series of market-based reforms in the 1990s, German retraining programs typically operate via “Bildungsgutschein” vouchers. Workers receive vouchers through registration with the Employment Agency, whose caseworkers conduct an initial interview and assessment ([Fitzenberger, Osikominu, and Paul 2023](#)). Both employed and unemployed workers can qualify, as caseworkers apply broad eligibility criteria which require only a risk of unemployment, missing qualifications, or poor chances of re-employment ([Kruppe 2008](#)). In some cases, caseworkers may require unemployed workers to participate in retraining as a condition of receiving unemployment benefits. In practice, participation is often largely voluntary, as the Employment Agency frequently tolerates dropouts ([Fitzenberger, Osikominu, and Paul 2023](#)).⁵ Subject to typically-mild caseworker-imposed restrictions on training content and sector, recipients can choose any certified private training provider ([Dauth 2020](#); [Fitzenberger, Osikominu, and Paul 2023](#)). The voucher then covers the direct costs of training as well as ancillary expenses such as transportation, and workers receive unemployment benefits for the duration of the program ([Bundesagentur für Arbeit 2021b](#)).

Table 1 summarizes the curricula of two such courses, both offered by TÜV Rheinland Akademie,

Measures History File (MTH/XMTH) portion of the SIAB. Note that the SIAB includes only retraining episodes starting in 2000 or later, so we restrict our worker panel to the post-2000 period.

⁵Employment Agency statistics indicate that 2.4% of *Sperrzeit* sanctions in 2010 were imposed due to refusal or dropout from career integration programs, which include retraining ([Bundesagentur für Arbeit 2011](#)).

a regional private training provider. The first, a short course on high-voltage fundamentals for electric vehicles, targets workers already employed in the automotive sector and covers the fundamentals of EV batteries and charging as well as safety practices over the course of one to four weeks. The second, a multi-month vehicle technology career entry program, serves workers retraining into the automotive sector from other industries. The three-to-six-month curriculum spans the geography of job opportunities, vehicle maintenance, diagnostics, new features at the frontier of automotive technology, and technical German vocabulary. Note that, like most German retraining courses, these examples encompass a mix of basic job search support and substantive, job-specific technical skills, and the aforementioned retraining vouchers would cover the full cost of either.

1.3 Four Retraining Facts

We now lay out four key facts to contextualize the role of retraining in the German labor market, and explain how they inform the model that will follow in Section 3. Supplemental Appendices A.1.1 and A.2 provide further details regarding the underlying data and variables.

Retraining is widespread. On average, 3.7% of German workers in our sample retrain each year, and, over a lifetime, a full 26% will have completed a retraining course at some point. Importantly, 74.9% of retraining spells coincide with non-employment: our model will allow employed and non-employed workers to face distinct retraining costs and thus generate different retraining rates by employment status.

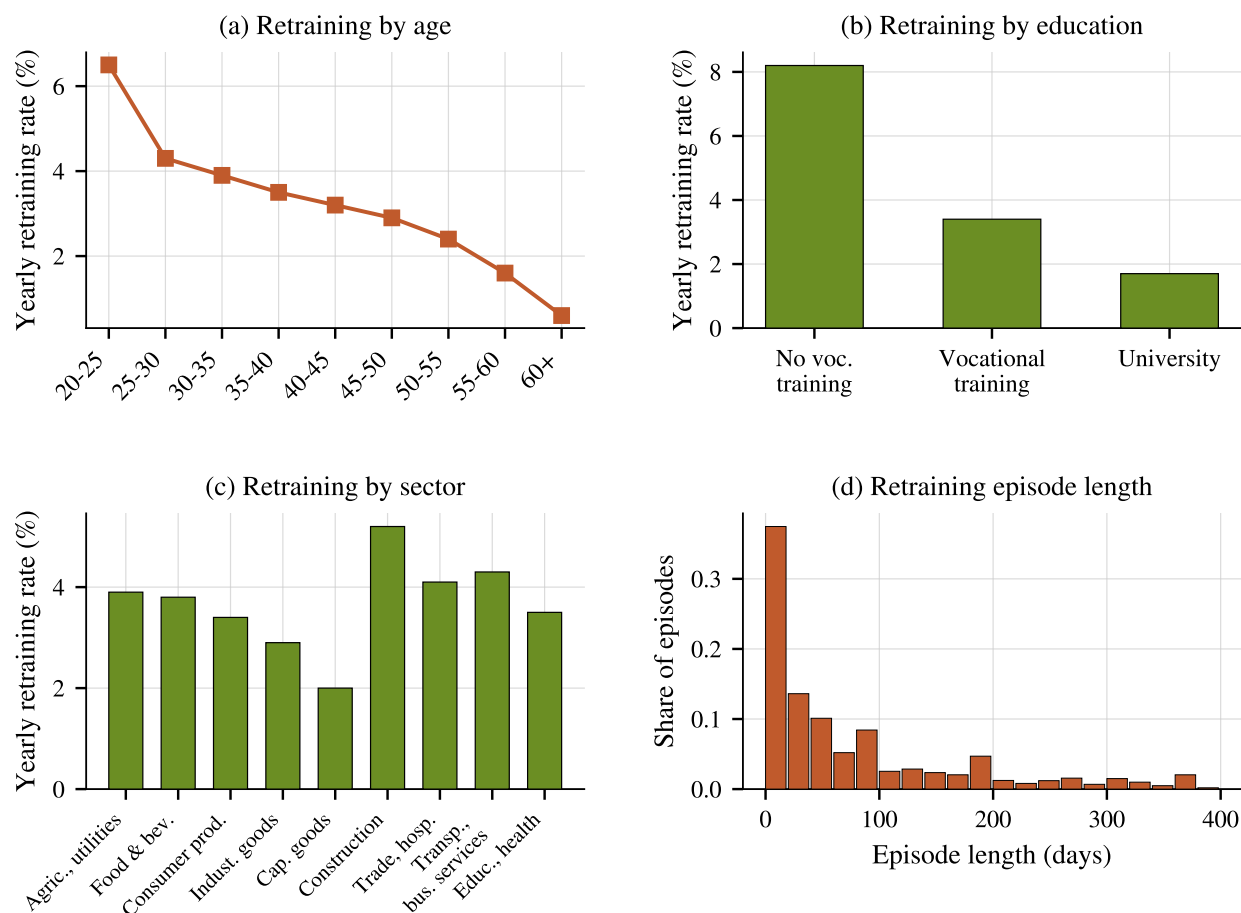
Retraining uptake is heterogeneous. A broad swath of the German labor force makes use of retraining, but with notable differences in uptake across workers. Panel (a) of Figure 1 shows that retraining rates decline monotonically with age, and Panel (b) shows that retraining is especially frequent among workers without vocational training or a degree. Panel (c) documents heterogeneity across broad sector groups. Informed by these patterns, our model will feature rich heterogeneity across ages and skill levels, and retraining rates will vary across sectors due to both differences in worker composition and endogenous differences in incentives to retrain.

Retraining courses are short on average, but vary widely in duration. Panel (d) of Figure 1 plots the distribution of retraining episode lengths, in days. The median retraining episode is relatively short at 41 days, with substantial variation around this average. While around a third of retraining episodes last less than 20 days, the distribution in Panel (d) features a long right tail: a full 29% of episodes exceed 100 days. Given the short duration of most retraining episodes, it seems unlikely that they would have permanent effects on human capital. Our model therefore focuses on relatively short-run effects of retraining, in particular its role in worker transitions across sectors.

Retraining is highly correlated with sector switching. The career-change course presented in Table 1 already hints at one potential use of retraining: switching sectors. We now explore more formally whether retrained workers tend to subsequently enter a new sector. We focus on workers initially employed in manufacturing, since these workers are the most exposed to trade shocks, and regress a dummy for switching sectors on a dummy for participation in subsidized retraining over the course of the decade 2000-2010. Table 2 reports our results.

In Column (1), we find that workers who retrain indeed turn out to switch sectors at a rate 38.9 percentage points above their non-retrained counterparts – a more than twofold increase relative to the non-retrained switching rate of 28%. This correlation might be partly driven by selection: workers who choose to retrain may be quite different than those who do not. To investigate this possibility, in Column (2), we control for gender, foreign nationality, age, education, and an East Germany indicator; and in Column (3), we further control for characteristics of workers’

Figure 1: Descriptive Statistics on Retraining



Notes: Figure displays yearly retraining rates by age (Panel a), education (Panel b), and broad sector groups (Panel c), along with the distribution of retraining episode lengths in days (Panel d), for workers in the SIAB (2000–2019). Episode lengths in Panel (d) are truncated at 400 days. Supplemental Appendices A.1.1 and A.2 detail the associated calculations.

Table 2: Retraining and Sector Switching

	Pr(Switch Sector)			
	(1)	(2)	(3)	(4)
Retraining dummy	0.389*** (0.005)	0.386*** (0.005)	0.366*** (0.005)	0.361*** (0.005)
Demographic Controls		✓	✓	✓
Initial Job Controls			✓	✓
Employment Control				✓
Observations	83,565	83,099	83,099	83,099

Notes: Robust standard errors in parentheses; *** indicates $p < 0.01$, ** $p < 0.05$, and * $p < 0.1$. OLS estimates of a regression of a dummy for switching out of the worker's initial 2000 sector during 2001–2010 on a dummy for participation in subsidized retraining during the same period, for workers initially employed in manufacturing. Columns add controls sequentially: Column (1) includes no controls; Column (2) adds demographic controls; Column (3) additionally adds initial job controls; Column (4) additionally adds the share of days employed during 2001–2010 as a control. Demographic controls include gender, age, education, foreign nationality, and an East Germany dummy. Initial, year-2000 job controls include tenure and firm size bins, manufacturing sector group indicators, and log earnings summed across all jobs.

base-year job including tenure, firm size, subsector, and log earnings. The strong correlation between retraining and sector switching persists even conditional on these characteristics.⁶ A related concern is that the correlation between retraining and switching might be driven by time spent in non-employment. In Column (4), we additionally control for the share of days a worker spends employed during the same decade over which we measure retraining and switching.⁷ Even among otherwise-similar workers who also spend the same amount of time employed, retrainees continue to switch sectors at higher rates.

The invariance of this estimated correlation to our battery of individual controls lessens the concern that it is purely the result of selection on worker heterogeneity. But it is still true that the decisions to switch sectors and retrain are very likely made jointly, with a worker's interest in joining a new sector influencing their decision to enroll in a retraining course. This simultaneity problem prevents us from directly inferring the causal effect of retraining from the regressions in Table 2. Instead, we will introduce a model of workers' decisions in Section 3 that disentangles these forces.

Finally, in Table 3, we investigate the sectors in which workers who retrain during unemployment find jobs, as compared to the unemployed who fail to retrain. Retrained workers flow disproportionately into transport, communications, and business services, which absorbs 37.9% of the retrained non-employed, as compared to 27.3% of the non-retrained. In contrast, retrained workers enter industrial goods manufacturing and construction at noticeably lower rates.

⁶Furthermore, Supplemental Appendix Table S2 shows that the same pattern holds across worker ages and skill levels.

⁷Of course, even if retraining were randomly assigned across workers, the regression in (4) would not have a causal interpretation as retraining may well affect the amount of time a worker spends in non-employment.

Table 3: Post-Non-Employment Sectors by Retraining Status

	Non-retrained	Retrained
Agriculture, mining, and utilities	0.046	0.030
Manufacture of food and beverages	0.022	0.021
Manufacture of consumer products	0.017	0.014
Manufacture of industrial goods	0.056	0.047
Manufacture of capital and consumer goods	0.049	0.035
Construction	0.106	0.078
Trade, hospitality, and other consumer services	0.227	0.222
Transport, communications, and business services	0.273	0.379
Education, health, and public administration	0.204	0.173

Notes: Table displays the share of non-employment spells which end with employment in each destination sector, separately for workers who participated in subsidized retraining during their spell of non-employment and for those who did not, over the years 2001–2010. Sectors correspond to the nine aggregate sectors delineated by [Dauth and Eppelsheimer \(2020\)](#).

2 Retraining, Import Competition, and the China Shock

We now show empirically that German workers use retraining to adjust to trade shocks: in our case, the post-2000 rise in imports from China to Germany. Following [Autor, Dorn, and Hanson \(2013\)](#), we refer to this as the China shock. We begin by documenting the adverse labor market consequences of exposure to the China shock and then explore the role of retraining as a margin of adjustment for workers.

2.1 Measuring the China Shock

We measure a worker’s exposure to the China shock using their initial sector of employment. We limit the sample to workers initially in manufacturing and then, following [Dauth, Findeisen, and Suedekum \(2021\)](#), we define the import exposure of a worker ω employed in the base year in sector s as

$$\text{import exposure}_s \equiv \frac{\Delta \text{net imports}_s}{\text{wage bill}_s},$$

where $\Delta \text{net imports}_s$ equals the 2001–2010 change in net imports from China to Germany. We focus not on a sector’s imports in isolation but on its net imports, i.e. imports minus exports, to capture the full impact of the China shock on labor demand, and we normalize net imports by the total base-year sectoral wage bill.⁸

With our exposure measure in hand, we estimate specifications for a series of labor market outcomes $y_{\omega s}$ for workers ω employed in manufacturing in sector s in the year 2000 as a function of sectoral import exposure,

$$y_{\omega s} = \beta \times \text{import exposure}_s + X_{\omega s} \gamma + \varepsilon_{\omega s}. \quad (1)$$

⁸Our sectors are a slightly aggregated version of ISIC Rev. 4 sectors, as detailed in Supplemental Appendix [A.1.1](#).

Table 4: Labor Market Effects of Chinese Import Exposure

<i>IV (2SLS)</i>	(1) Share days employed	(2) Share days in manufacturing	(3) Earnings growth	(4) Sector switching
Sectoral import exposure	-2.340*** (0.701)	-4.880*** (1.610)	-2.748*** (0.930)	4.230* (2.380)
Demographic controls	✓	✓	✓	✓
Initial job controls	✓	✓	✓	✓
Observations	83,099	83,099	83,099	83,099
Mean of dependent variable	76.1	64.9	80.965	21.8
First-stage <i>F</i> -statistic	237.63	237.63	237.63	237.63

Notes: Robust standard errors in parentheses, clustered at the sector level; *** indicates $p < 0.01$, ** $p < 0.05$, and * $p < 0.1$. We measure the share of days employed and share of days employed in manufacturing (unconditional on employment) in 2010, earnings growth as 100 times total 2010 earnings divided by those in 2000, and sector switching as a dummy for any employment in 2010 outside of a worker's initial sector. Shares and probabilities are multiplied by 100; these outcomes can be interpreted as percentage points.

Controls in $X_{\omega s}$ include demographic characteristics measured in 2000—age bins, gender, education, foreign nationality, and an East Germany dummy—as well as characteristics of their primary base-year, or *initial*, job: log earnings summed across all jobs, tenure and firm size bins, and broad manufacturing sector group indicators. We relegate further details regarding the construction of our outcomes, import exposure measure, and controls to Supplemental Appendix A.2.

The threat to identification of the coefficient of interest, β , in (1) is domestic supply and demand shocks correlated with changes in sectoral import exposure. We therefore follow Autor, Dorn, and Hanson (2013) and instrument for *import exposure_s* with an analogous measure which instead uses net imports from China to other high-income countries.⁹

2.2 Labor Market Effects of the China Shock

Table 4 presents 2SLS estimates of the labor market effects of Chinese import exposure between 2000 and 2010. We measure four outcomes at the end of the decade: the share of days employed, in percentage points (Column 1), the share of days employed in manufacturing, unconditional on employment (Column 2), cumulative earnings growth, defined as 100 times end-of-decade divided by base-year earnings (Column 3), and a dummy for switching out of the worker's initial sector (Column 4).

A one-standard-deviation, or 0.67-unit, increase in sectoral import exposure reduces total employment by 1.6 percentage points, or 2.1% of the mean (Column 1), and manufacturing employment by 3.3 percentage points, or 5.1% of the mean (Column 2). These employment declines

⁹As in Dauth, Findeisen, and Suedekum (2021), these are the UK, Sweden, Norway, Canada, Australia, New Zealand, Japan, and Singapore.

translate directly into lower earnings growth: the same shock reduces cumulative earnings growth by 1.9 percentage points (Column 3). Taken together, these results confirm the adverse effects of trade exposure on the German labor market established by [Dauth, Findeisen, and Suedekum \(2021\)](#). To begin to understand how trade-exposed workers adjust, we look for evidence of sectoral reallocation. The same shock raises the probability of switching sectors by 2.8 percentage points, or 12.8% of the mean (Column 4). In Supplemental Appendix Table [S3](#), we show that these results are robust to measuring import exposure and sector switching at the finer 3-digit sector level.

2.3 The China Shock and Retraining

We now provide direct evidence of a new margin of adjustment to trade shocks: retraining. In Column (1) of Table [5](#), we estimate equation [\(1\)](#) with a dummy variable $y_{\omega s}$ equal to one for workers who retrain at any point during the full decade 2001–2010, initially without controls. In Column (2), we add controls for demographic and initial-job characteristics. The coefficient in our preferred specification in Column (2), significant at the 5% level, implies that a one-standard-deviation increase in import exposure increases the probability that a worker retrain by 1.1 percentage points — an 8.1% increase on the mean 10-year retraining rate of 13.5%. In Column (3) we estimate a logit specification using a control function approach and obtain similar results: a one-standard-deviation increase in sectoral import exposure increases a worker’s relative odds of retraining by around 6.5%. Finally, Column (4) explores the intensive margin by focusing on the number of days spent retraining. We again find a positive, but in this case statistically insignificant, effect of import exposure on retraining.¹⁰

In principle, increased retraining among more exposed workers could reflect two forces: changes in the distribution of workers across employment and non-employment, and changes in workers’ propensity to retrain conditional on their employment status. We now turn to a decomposition that formalizes this idea. Let $\Pr(R)$ denote the unconditional retraining probability across workers, $\Pr(N)$ the probability of non-employment, and $\Pr(R|N)$ and $\Pr(R|E)$, the retraining probabilities conditional on non-employment and employment, respectively. Then, we can decompose the total effect of import exposure ΔM on retraining rates as

$$\frac{\partial \Pr(R)}{\partial \Delta M} = (\Pr(R|N) - \Pr(R|E)) \frac{\partial \Pr(N)}{\partial \Delta M} + \Pr(N) \frac{\partial \Pr(R|N)}{\partial \Delta M} + (1 - \Pr(N)) \frac{\partial \Pr(R|E)}{\partial \Delta M}. \quad (2)$$

The first term captures the mechanical shift of workers into non-employment, where baseline retraining rates are higher; the latter two terms capture changes in retraining rates conditional on employment status.

In Supplemental Appendix [A.5](#), we estimate each quantity on the right-hand side of [\(2\)](#) and find that both forces play a role. Exposure to import competition shifts workers into non-employment, and, combined with differences in baseline retraining rates between employed and non-employed

¹⁰As with the labor market outcomes above, we find quantitatively very similar results using a finer 3-digit sector classification in Supplemental Appendix Table [S3](#).

Table 5: Chinese Import Exposure and Retraining

<i>Outcome variable</i> <i>Specification</i>	(1) Pr(retrain) Linear probability model	(2) Pr(retrain) 2SLS	(3) Pr(retrain) Logit	(4) Days retraining 2SLS
Sectoral import exposure	4.300*** (1.100)	1.564** (0.697)	0.094*** (0.023)	2.338 (1.728)
Demographic controls		✓	✓	✓
Initial job controls		✓	✓	✓
Observations	83,565	83,099	83,099	83,099
Mean of dependent variable	13.5	13.5	0.135	17.030
First-stage <i>F</i> -statistic	181.17	237.63		237.63

Notes: Robust standard errors in parentheses, clustered at the sector level, in Columns 1–2 and 4; standard errors in Column 3 bootstrapped. *** indicates $p < 0.01$, ** $p < 0.05$, and * $p < 0.1$. The retraining outcome in Columns 1–3 is a dummy for participation in subsidized retraining during 2001–2010; Column 4 measures the number of days of retraining. Column 3 estimates a logit control function specification. Probabilities are multiplied by 100; these outcomes can be interpreted as percentage points.

workers, this shift accounts for 32.9% of the effect in Column (2) of Table 5. But most of the effect – 67.1% – is driven by changes in workers’ decisions conditional on their employment status. We interpret this breakdown as evidence that workers do not simply retrain as a mechanical result of unemployment or employment agency requirements, but instead actively choose to participate in these programs in response to real economic incentives.

2.4 Taking Stock

Two findings stand out from our empirical analysis. First, in the cross-section, retraining and intersectoral mobility are tightly linked: workers who retrain are more than twice as likely to switch sectors than those who do not (Table 2). Second, in response to the China shock, both margins move together—a one standard deviation increase in import competition raises sector switching by 12.8% and retraining by 8.1%, and only a minority of the latter effect stems from mechanical shifts into non-employment. These patterns point towards a complementarity between retraining and intersectoral mobility — and hence a tight link between the availability of retraining and workers’ ability to smooth trade shocks.

At the same time, the joint nature of the retraining and sector-switching decisions means that we cannot pin down the true direction of causality from these patterns alone. We now develop a model of workers’ retraining and sector switching decisions that will allow us to address the endogeneity of retraining and to explore the interaction of trade, intersectoral mobility, and retraining in equilibrium.

3 A Theory of Trade and Retraining

We now introduce a model of retraining and intersectoral mobility which allows us to pin down the complementarity between these two decisions and thus quantify how retraining helps workers adjust to trade shocks in equilibrium. We begin with a stylized model which highlights the effect of retraining on the labor market dynamics and welfare changes induced by a trade shock. We then add worker heterogeneity and embed our model in a quantitative, general equilibrium model of international trade.

3.1 Simple Model

Environment

Time is discrete and indexed by t . We study a single small open economy with tradable sectors $i = 1, \dots, N$, so that world prices P_{it} are exogenous. Production in each sector has constant returns to scale, uses only labor, and is carried out by a mass of identical, perfectly competitive firms with productivities normalized to one. As a result, nominal wages $\tilde{w}_{it}$ equal world prices. The economy is populated by a unit mass of infinitely-lived workers who discount the future with a discount factor β .

The Worker's Problem

Workers have log utility over their hand-to-mouth consumption bundle in each period. The latter consists of goods from different sectors, over which workers have Cobb-Douglas preferences with weights α_i . Consider a single worker who starts period t in sector i . She begins the period by supplying one unit of labor in exchange for a nominal wage $\tilde{w}_{it}$ and then consumes her real wage, which, under Cobb-Douglas preferences, is given by

$$w_{it} \equiv \tilde{w}_{it} P_t^{-1}, \quad P_t = \prod_{j=1}^N P_{jt}^{\alpha_j}, \quad (3)$$

where P_t is the aggregate price index in period t . We denote the expected lifetime utility of a worker in sector i in period t by v_{it} .

At the end of the period, the worker must make two decisions: whether or not to switch out of her current sector i into some other sector j , and whether or not to retrain. To capture the interaction of these two decisions, we combine the model of complementary choices in [Gentzkow \(2007\)](#) with the model of labor mobility developed by [Artuç, Chaudhuri, and McLaren \(2010\)](#). In particular, workers draw idiosyncratic preference shocks ϵ_{ij} for each destination j (including the current sector), scaled by a dispersion parameter ν . Workers who switch sectors pay an additive deterministic switching cost C .

Retraining, in turn, reshapes the net payoff to switching sectors. In particular, a worker who retrain faces switching costs of $C - \Gamma$. Thus, when $\Gamma > 0$, retraining lowers the cost of

switching sectors, and retraining and switching are complements: we therefore refer to Γ as the *complementarity effect*.¹¹ This specification allows us to parsimoniously capture the benefits of the range of activation services, placement assistance, and technical instruction offered by retraining programs. Retrained workers also experience an idiosyncratic benefit shock η scaled by a parameter σ and pay a deterministic cost F .

The payoffs to the worker from these different decisions thus read as follows:

$$\begin{aligned}
\text{Do not switch \& do not retrain} &\implies \beta v_{it+1} + v\epsilon_{ii} \\
\text{Switch to sector } j \text{ \& do not retrain} &\implies \beta v_{jt+1} + v\epsilon_{ij} - C \\
\text{Do not switch \& retrain} &\implies \beta v_{it+1} + v\epsilon_{ii} + (\sigma\eta - F) \\
\text{Switch to sector } j \text{ \& retrain} &\implies \beta v_{jt+1} + v\epsilon_{ij} + (\sigma\eta - F) - (C - \Gamma).
\end{aligned}$$

These payoffs underscore how the complementarity effect Γ shapes the interaction between retraining and intersectoral mobility. If $\Gamma > 0$, the two are complements – retraining primarily helps workers acquire skills useful outside their current sector. As we will see below, the sign and magnitude of Γ are crucial for the interaction of retraining and trade shocks.

As is standard in the literature, we assume that the sectoral preference shocks ϵ_{ij} are drawn from a Type 1 Extreme Value distribution with mean zero and scale parameter normalized to one. For tractability, we also assume that the idiosyncratic retraining benefit η is drawn from a logistic distribution with mean zero and scale one. We summarize the problem facing the worker with a Bellman equation,

$$v_{it} = \log w_{it} + U_{it}, \quad (4)$$

where U_{it} is the inclusive value,

$$U_{it} = \mathbb{E} \left[\max_j \{ \beta v_{jt+1} + v\epsilon_{ij} - C \mathbb{1}\{j \neq i\} + \max\{ \Gamma \mathbb{1}\{j \neq i\} + \sigma\eta - F, 0 \} \} \right], \quad (5)$$

and the expectation is taken over the idiosyncratic shocks ϵ_{ij} and η . Note that the indicator $\mathbb{1}\{j \neq i\}$ ensures that switching costs C and the complementarity effect Γ apply only when the worker changes sector.

Equilibrium

Given an initial distribution of labor across sectors, an equilibrium of the model is: a sequence of worker values v_{it} which solve the Bellman equation (4); given these worker values, a sequence of switching probabilities μ_{ijt} – denoting the probability a worker in sector i moves to sector j – and retraining rates ρ_{it} implied by (4); and finally a sequence of labor allocations l_{it} across sectors

¹¹Although it is natural to expect that retraining and switching are complements, so that $\Gamma > 0$, our model does allow for the possibility that $\Gamma < 0$. Our calibration procedure in Section 4 does not *a priori* restrict the sign of Γ .

satisfying,

$$l_{it+1} = \sum_{j=1}^N \mu_{jit} l_{jt}.$$

3.2 Analytical Results

We now leverage this stylized model to analytically characterize the labor market and welfare effects of retraining following a trade shock. Recall that, in our setup, real wages w_{it} in (3) depend only on exogenous parameters and world prices P_{it} . We therefore consider a one-time, permanent shock to world prices and thus real wages, which we loosely interpret as an import competition shock. Following the shock, real wages, and hence worker values and choice probabilities, are constant over time. We therefore drop the t subscripts where doing so does not cause confusion. Proofs of all the results in this section can be found in Appendix B.

Retraining and Worker Choices

We start by characterizing how retraining alters worker choices. While the simultaneity of workers' retraining and sector switching decisions will later allow the model to capture an important endogeneity problem in the data, this structure generates intractable worker choice probabilities. We therefore take a first order approximation around $\Gamma = 0$; close to these parameter values, the model can be solved by hand but nonetheless captures the interdependence of retraining and switching.

Theorem 1. *To a first order approximation around $\Gamma = 0$, the probability that a worker initially in sector i switches to sector j , relative to the probability of remaining in i , is,*

$$\log \left(\frac{\mu_{ij}}{\mu_{ii}} \right) = \left(\frac{\beta}{\nu} \right) (v_j - v_i) - \left(\frac{C - \Gamma \bar{\rho}}{\nu} \right), \quad (6)$$

and the retraining rate is,

$$\rho_i = \bar{\rho} + \bar{\rho}(1 - \bar{\rho}) \left(\frac{\Gamma}{\sigma} \right) (1 - \bar{\mu}_{ii}). \quad (7)$$

In these expressions, $\bar{\mu}_{ii}$ is the probability of remaining in sector i , evaluated at $\Gamma = 0$, and $\bar{\rho}$ is the corresponding retraining rate, common across sectors under $\Gamma = 0$.

Theorem 1 clarifies the role of retraining in our model. The probability that a worker switches sectors takes the standard extreme-value form in (6), but with switching costs shifted by $-\Gamma \bar{\rho}$. If switching and retraining are complements, then switching costs effectively fall as retraining rises, whereas, if they are substitutes, the opposite is true. In either case, the key complementarity parameter Γ , scaled by the retraining benefit dispersion ν , governs the effect of retraining on switching. Finally, (7) demonstrates a natural link between retraining rates and mobility across sectors, with the sign of this relationship again determined by Γ .

Retraining and Welfare

How does retraining alter the distribution of welfare gains and losses from trade? Consider a vector of small, permanent shocks to world prices $d \log P_i$, starting from a symmetric steady state at $t = 0$; workers learn of the shock between $t = 0$ and $t = 1$ and choose sectors for $t = 1$ accordingly.¹² We measure the welfare change of workers initially in sector i as equivalent variation, denoted EV_i and expressed as a percentage of consumption. We show in Appendix A that in our model this is given by $EV_i = dU_{i0}$, where U_{i0} is the inclusive value from (5).

In this simple setting, retraining shapes welfare changes EV_i entirely through workers' baseline mobility level across sectors, which, for notational convenience, we define as $\tilde{m} = \frac{N}{N-1}(1 - \mu_{ii})$. Rearranging the switching probabilities in Theorem 1 delivers an intuitive expression for baseline mobility in a symmetric-world-price steady state:

$$\tilde{m} = \frac{N}{e^{(C-\Gamma\bar{\rho})/\nu} + N - 1}. \quad (8)$$

Whenever $\Gamma > 0$, increases in the retraining rate $\bar{\rho}$ raise baseline mobility; to emphasize this dependence we henceforth write $\tilde{m}(\bar{\rho})$.

Our next result uses $\tilde{m}(\bar{\rho})$ to decompose welfare changes into a direct wage effect and an option value term.

Theorem 2. *Assume the economy is initially in a symmetric steady state with $P_i = P_j$ for all sectors i and j and that, between $t = 0$ and $t = 1$, world prices change permanently by $d \log P_i$. Then:*

(i) *Changes in real wages are given by*

$$d \log w_i = d \log P_i - \sum_j \alpha_j d \log P_j. \quad (9)$$

(ii) *The employment-weighted average change in welfare is*

$$\overline{EV} = \left(\frac{\beta}{1 - \beta} \right) \sum_i (l_i - \alpha_i) d \log P_i. \quad (10)$$

(iii) *The change in the welfare of workers initially in sector i is*

$$EV_i = \left(\frac{\beta}{1 - \beta} \right) d \log w_i + \lambda(\bar{\rho}) \left[\overline{EV} - \left(\frac{\beta}{1 - \beta} \right) d \log w_i \right], \quad (11)$$

where

$$\lambda(\bar{\rho}) \equiv \frac{\tilde{m}(\bar{\rho})}{1 - \beta + \beta \tilde{m}(\bar{\rho})} \quad (12)$$

captures the importance of the option value the worker derives from being able to switch sectors in the future.

¹²No results hinge on this timing assumption, but it matches the setup we will use for our calibration and counterfactual exercises.

Crucially, retraining does not change how this trade shock affects real wages or *average* welfare. Real wages depend only on relative price changes, while the average welfare effect depends on the correlation between the price changes and employment shares l_i versus consumption shares α_i .

Instead, retraining – through its effect on mobility $\tilde{m}(\bar{\rho})$ – governs the distribution of welfare changes across workers. Equation (11) decomposes these sectoral welfare changes into a direct effect – the real wage change facing a completely immobile worker – captured by the first term, and, in the second term, an option value, which accounts for the worker’s ability to move to a new sector in future. The two are negatively correlated: workers hit hardest by the shock are precisely those for whom switching is most attractive. To see the role of retraining clearly, assume that workers are relatively patient, so that $\beta \rightarrow 1$, and that the approximation in Theorem 1 holds. Then, the *relative* standard deviation of shock-induced welfare changes EV_i across workers in an economy with retraining rate $\bar{\rho}$ versus one with no retraining is

$$\Delta \text{Std}(EV_i) \simeq - \left(\frac{\Gamma \bar{\rho}}{\nu} \right) \times \left[N^{-1} \exp \left(\frac{C}{\nu} \right) \right] \times \text{Std} \left(\frac{\beta}{1-\beta} d \log w_i \right). \quad (13)$$

This expression highlights a key insight: whenever retraining complements sector-switching, i.e. $\Gamma > 0$, a higher retraining rate $\bar{\rho}$ shrinks this standard deviation and compresses the distributional effects of trade shocks.

Retraining and Labor Market Dynamics

A large literature has documented sluggish labor market adjustment to international trade shocks. Our final analytical result considers how the availability of retraining might speed or slow these transitions. To derive this set of results, we assume that retraining and switching are complements, i.e. $\Gamma > 0$. In consequence, reductions in the cost of retraining or increases in its effectiveness will raise baseline mobility $\tilde{m}(\bar{\rho})$.

Theorem 3. *Under the first-order approximation in Γ introduced in Theorem 1 and following the shock to world prices described in Theorem 2:*

(i) *The change in employment in sector i in the short run ($t = 1$) is*

$$d \log l_{i1} = \tilde{m}(\bar{\rho}) [2 - \tilde{m}(\bar{\rho})] \left(\frac{1 - \beta \lambda(\bar{\rho})}{\nu} \right) \left[\left(\frac{\beta}{1 - \beta} \right) d \log w_i - \overline{EV} \right]. \quad (14)$$

Thus, a higher mobility level $\tilde{m}(\bar{\rho})$ has ambiguous effects on the magnitude of short-run employment reallocation in general, but for sufficiently low mobility $\tilde{m}(\bar{\rho})$, increases in $\tilde{m}(\bar{\rho})$ result in more short-run reallocation.

(ii) *The long-run change in employment as $t \rightarrow \infty$ is*

$$d \log l_{i\infty} = [2 - \tilde{m}(\bar{\rho})] \left(\frac{1 - \beta \lambda(\bar{\rho})}{\nu} \right) \left[\left(\frac{\beta}{1 - \beta} \right) d \log w_i - \overline{EV} \right]. \quad (15)$$

A higher mobility level $\tilde{m}(\bar{\rho})$ always decreases the magnitude of long-run employment reallocation.

(iii) The speed of convergence, measured by the fraction of the long-run adjustment complete at time t , is

$$\frac{d \log l_{it}}{d \log l_{i\infty}} = 1 - [1 - \tilde{m}(\bar{\rho})]^t. \quad (16)$$

Thus, a fraction $\tilde{m}(\bar{\rho})$ of long-run adjustment is completed one period after the shock, and a higher $\tilde{m}(\bar{\rho})$ always speeds up convergence.

Theorem 3 decomposes the effect of retraining on employment reallocation into two forces. The first runs through welfare. From Theorem 2, a higher baseline retraining rate $\bar{\rho}$ compresses the distribution of welfare changes induced by trade shocks. This compression – captured by the, under $\Gamma > 0$, increasing function $\lambda(\bar{\rho})$ in (14) and (15) – weakens workers’ incentives to move and shrinks the employment response at every horizon. In the short run, an additional countervailing effect runs through baseline mobility $\tilde{m}(\bar{\rho})$, which has the intuitive effect of magnifying the employment response to a given shock. For the empirically relevant case of low baseline mobility, the second force dominates, and retraining magnifies short-run employment responses. In the long run, in contrast, retraining unambiguously shrinks the magnitude of the employment response through the aforementioned welfare compression channel.

This long-run dampening effect of retraining may at first appear counterintuitive. Indeed, retraining lowers switching costs, so it is tempting to conclude that it must raise long-run reallocation. However, this logic conflates two distinct objects. As in the classic [Artuç, Chaudhuri, and McLaren \(2010\)](#) framework, our model features gross worker flows in excess of net flows. Retraining indeed raises gross flows in both directions, but in the long run, since workers can easily exit any unattractive sector, net reallocation across sectors falls.

Finally, while retraining shrinks steady-state employment changes, (16) shows that it speeds the transition. The fraction of long-run adjustment completed by period t equals $1 - (1 - \tilde{m}(\bar{\rho}))^t$: a share $\tilde{m}(\bar{\rho})$ of the entire transition occurs in the first period alone. The same baseline mobility that dampens the long-run response thus also governs the speed of convergence to it. An economy with high retraining rates reaches its new steady state quickly, but this new steady state involves less reallocation; an economy without retraining moves slowly towards a more distant steady state.

Three key insights thus emerge from the simple model. First, retraining effectively reduces the costs of moving across sectors and raises worker mobility. Second, when an economy is hit by trade shock, retraining compresses the distribution of welfare changes across workers without changing its average effects. And third, economies with retraining see smaller shifts in employment across sectors in the long-run, but the employment reallocation that does occur is more front-loaded.

Retraining and Human Capital

Our model treats switching and retraining costs as non-pecuniary shifters to utility, following [Artuç, Chaudhuri, and McLaren \(2010\)](#) and [Caliendo, Dvorkin, and Parro \(2019\)](#). A parallel literature,

including Dix-Carneiro (2014) and Traiberman (2019), instead emphasizes sector-specific human capital. Supplemental Appendix B.1 extends the model in this direction, allowing retraining to alter a worker’s stock of human capital. We show that this extended model is isomorphic to our baseline framework, so the above predictions of the model for welfare and transition dynamics are unchanged. We therefore do not need to take a stance on whether retraining works through non-pecuniary costs or human capital – only whether it complements or substitutes for switching.¹³

3.3 Quantitative Model

We now enrich the model for quantitative analysis. Motivated by the heterogeneity in retraining rates in the data in Section 1, we model workers of different skill levels and ages. We also allow for a non-employment option. Finally, to connect our model to the evidence from Section 2 on the effects of import competition, we embed our full labor market framework for a single “home” country within a quantitative, general equilibrium model of international trade.

Worker Heterogeneity

We allow for worker heterogeneity along two dimensions. First, workers have an exogenous skill level, indexed by k , which remains fixed over time. We denote the share of each skill group in the labor force by λ^k . Second, we capture heterogeneity in age, indexed by a , with a simple overlapping generations (OLG) structure. Workers are born at an exogenous rate δ as new entrants, $a = e$. New entrants choose a sector and then immediately become young workers, $a = y$. At some exogenous rate χ young workers become old, $a = o$. In turn old workers retire at rate ζ and then exit the labor market. We assume a constant population, which corresponds to $\delta = \zeta\chi(\zeta + \chi)^{-1}$.

Worker types thus correspond to skill-age tuples. A worker of type ka supplies h_i^{ka} units of effective labor when working in sector i and faces switching costs C_{ij}^{ka} and preference shocks ϵ_{ij} if she chooses to move from sector i to sector j . As in the simple model, we assume $C_{ii}^{ka} = 0$ but otherwise allow switching costs to vary flexibly across origin-destination pairs. We write new entrants’ entry costs to sector j as C_j^{ke} . We also allow for variation across worker types in the costs and benefits of retraining, denoted by F^{ka} and Γ^{ka} , respectively. Note, finally, that we assume that new entrants cannot retrain before choosing a sector.

Non-Employment

We augment our model with a non-employment option, indexed by $i = 0$. While in non-employment, workers enjoy a type-specific flow of utility b^{ka} . We model entry into and exit out of this additional “sector” in the same way as any other, and workers in non-employment also have the option to retrain. However, their retraining costs may differ from those of employed

¹³Of course, a model in which retraining operates through human capital rather than non-pecuniary switching costs has different implications for earnings. In Section 4 we identify the effects of retraining without using information on earnings and so sidestep this issue.

workers: we denote the costs facing non-employed workers by F_0^{ka} and those facing employed workers by F_1^{ka} . Finally, we assume that retraining only affects the cost of entering productive sectors, i.e., excluding non-employment.

Bellman Equations

We summarize these various model ingredients with a system of Bellman equations. At time t , a new entrant from skill group k draws preference shocks again denoted by ϵ and chooses an initial sector j to solve

$$\max_j \{ \beta v_{jt+1}^{ky} + v\epsilon_j - C_j^{ke} \}, \quad (17)$$

where v_{jt+1}^{ky} is the value of starting the next period as a young worker in sector j .

An *employed* worker of type ka in a productive sector $i \neq 0$ derives utility from the log of her consumption $w_{it}h_i^{ka}$ plus an exogenous sectoral utility shifter ψ_i^{ka} .¹⁴ Her value function satisfies

$$v_{it}^{ky} = \log(w_{it}h_i^{ky}) + \psi_i^{ky} + (1 - \chi)U_{it}^{ky} + \chi U_{it}^{ko}, \quad (18)$$

$$v_{it}^{ko} = \log(w_{it}h_i^{ko}) + \psi_i^{ko} + (1 - \zeta)U_{it}^{ko}, \quad (19)$$

if she is young or old, respectively. The continuation value U_{it}^{ka} is given by

$$U_{it}^{ka} = \mathbb{E} \left[\max_j \left\{ \beta v_{jt+1}^{ka} + v\epsilon_{ij} - C_{ij}^{ka} + \max \{ \Gamma^{ka} \mathbb{1}\{j \neq i\} \mathbb{1}\{j \neq 0\} + \sigma\eta - F_1^{ka}, 0 \} \right\} \right], \quad (20)$$

where F_1^{ka} is the cost of retraining for employed workers and Γ^{ka} is the retraining-switching complementarity parameter for workers of type ka . Notice that the worker only gains from this complementarity between switching and retraining if she moves to a new productive sector, i.e., a sector, other than non-employment, different from her initial sector i . The problem facing *non-employed* workers is almost identical, except that (i) their flow payoffs equal b^{ka} in place of log real earnings, and (ii) they face retraining costs F_0^{ka} .

We denote the resulting type-specific sector-transition and retraining probabilities by μ_{ijt}^{ka} and ρ_{ijt}^{ka} , respectively. Given a distribution l_{it}^{ka} of worker types over sectors at time t , employment evolves according to,

$$l_{it+1}^{ky} = (1 - \chi) \sum_{j=0}^N \mu_{jit}^{ky} l_{jt}^{ky} + \delta \lambda^k \mu_{it}^{ke}, \quad l_{it+1}^{ko} = \sum_{j=0}^N \mu_{jit}^{ko} \left((1 - \zeta) l_{jt}^{ko} + \chi l_{jt}^{ky} \right) \quad (21)$$

for each skill group k . Finally, total effective labor supply in sector i is,

$$L_{it} = \sum_k \sum_a l_{it}^{ka} h_i^{ka}. \quad (22)$$

¹⁴Technically speaking, we could dispense with the flow utility shifters ψ_i^{ka} and absorb them into switching costs. Including them allows us to impose the natural normalization that $C_{j0}^{ka} = 0$, i.e., entering non-employment is always costless.

Other Countries

We model labor supply in the country of interest – Germany, in our application – as described in the preceding sections. In other countries, indexed by n , we adopt a simpler model: a mass of L_n homogeneous workers each supply one unit of labor and are perfectly mobile across sectors. Wages w_{nt} in each country n in each period t are therefore identical across sectors. We do not allow for non-employment in other countries.

International Trade

The international trade side of our model is standard and closely follows [Caliendo and Parro \(2015\)](#).¹⁵ We therefore keep our exposition brief and, since this block of the model is entirely static, drop time subscripts in what follows.

Firms in sector i in country n produce a continuum of varieties of goods under constant returns to scale using a nested Cobb-Douglas bundle of labor, structures, and intermediate inputs from other sectors. The upper nest combines a composite “value-added” factor of production, which itself is a Cobb-Douglas bundle of labor and structures, and a Cobb-Douglas bundle of intermediate inputs. We assume that structures are in fixed aggregate supply S_n within a country but perfectly mobile across sectors, implying a common structures rental rate r_n . The resulting unit costs, expressed in logs, are,

$$\log c_{in} = \gamma_{in}((1 - \phi_{in}) \log w_{in} + \phi_{in} \log r_n) + (1 - \gamma_{in}) \sum_{j=1}^N \xi_{ijn} \log P_{jn}, \quad (23)$$

where P_{jn} is the price index for sector j goods in country n , γ_{in} is the value added share in i , ϕ_{in} is the share of structures in value added in i , and ξ_{ijn} is the share of sector j in the intermediate inputs purchased by sector i in n . Given unit costs c_{in} , trade between countries in sector i is modeled exactly as in [Eaton and Kortum \(2002\)](#). The resulting shares π_{inm} of country n in country m 's purchases of sector i goods and price indices P_{im} for sector i in country m are,

$$\pi_{inm} = \frac{(\tau_{inm} c_{in})^{-\theta_i} T_{in}}{P_{im}^{-\theta_i}}, \quad P_{im} = \left(\sum_n (\tau_{inm} c_{in})^{-\theta_i} T_{in} \right)^{-\frac{1}{\theta_i}}. \quad (24)$$

Here T_{in} and θ_i are the scale and shape parameters of the Frechet distribution from which the productivities of producers of different varieties in sector i in country n are drawn: a country with a high T_{in} is relatively productive in sector i on average. Lastly, τ_{inm} represents an iceberg cost of shipping sector i goods from n to m .

Consumers in each country have Cobb-Douglas preferences over different sectors with weights α_{in} . As in [Dekle, Eaton, and Kortum \(2007\)](#), we allow for exogenous trade deficits D_n that create a

¹⁵Note that we dispense with [Caliendo and Parro \(2015\)](#)'s spatial dimension.

wedge between final expenditures and total value added. Market clearing then requires:

$$X_{in} = \alpha_{in} Y_n + \sum_{j=1}^N (1 - \gamma_{jn}) \xi_{jin} R_{jn}, \quad (25)$$

$$Y_n = D_n + \sum_{i=1}^N \left(\frac{w_{in} L_{in}}{1 - \phi_{in}} \right), \quad (26)$$

$$R_{in} = \sum_m \pi_{im} X_{im}, \quad (27)$$

$$w_{in} L_{in} = \gamma_{in} (1 - \phi_{in}) R_{in}, \quad (28)$$

$$r_n S_n = \sum_{i=1}^N \gamma_{in} \phi_{in} R_{in}, \quad (29)$$

where X_{in} and R_{in} denote total expenditures and revenues in sector i in country n , Y_n is total final expenditure, and, recall, S_n is the exogenous supply of structures. In Germany, L_{in} is determined by the labor market module of our model and (28) pins down sector-specific wages. In other countries, labor mobility implies a single wage w_n , determined by aggregating (28) over sectors.

Equilibrium

Following [Caliendo, Dvorkin, and Parro \(2019\)](#), we define the *fundamentals* of an economy as its switching costs C , retraining costs F and complementarities Γ , type-specific units of effective labor h and non-employment utilities b , as well as its input-output coefficients γ and ξ , its trade costs τ , its productivities T , its Cobb-Douglas consumption shares α and its exogenous deficits D . Given a set of fundamentals and a distribution of effective units of labor across sectors in Germany, we then define a *static equilibrium* as wages w , rental rates r , prices P , unit costs c , trade shares π , expenditures X and revenues R that solve (23) – (29). Finally, given an initial distribution of labor across sectors in Germany and a (possibly time-varying) sequence of fundamentals, a *dynamic equilibrium* is a sequence of static equilibria for each period t plus corresponding sequences of worker values v , sector transition probabilities μ , retraining decisions ρ , and labor supplies l satisfying (18) – (22) and consistent with the static equilibria and fundamentals.

4 Calibration

The model requires four groups of parameters. We calibrate the first group, reported in Table 6, externally: the trade elasticities θ_i , the discount factor β , the parameters of the OLG structure, and the population shares of each skill group. The second group consists of trade-side fundamentals – productivities, trade costs, input-output coefficients, and so on – which we absorb and thus avoid the need to calibrate entirely by solving the model in changes following [Dekle, Eaton, and Kortum \(2007\)](#). We recover the third group, namely the switching costs C_{ij}^{ka} , from observed worker flows. The fourth group is a vector $\Theta = \{\nu, \sigma, \{\Gamma^{ka}, F_0^{ka}, F_1^{ka}\}_{ka}\}$ of twenty parameters governing

preferences over sectors, retraining costs, and the complementarity between retraining and mobility. We calibrate Θ by matching the empirical labor market effects of the China shock from Section 2, along with other auxiliary moments.

To operationalize our model empirically, we must define time periods and worker types. We take a period to be five years; this longer timespan allows the model to capture the idea that the effects of retraining may persist over a number of years. As for worker types, we measure skill as a worker’s education level – no higher education ($k = L$), vocational training (M), or a university degree (H) – upon entering the labor market. Young workers ($a = y$) are those aged 18-44, while old workers (o) are those aged 45-65.

4.1 Hat Algebra

Our calibration procedure starts from a baseline equilibrium and solves the model forward given a sequence of shocks to fundamentals. Given data on the baseline equilibrium and a candidate parameter vector Θ , the model’s static equilibrium can be solved in changes via the standard hat algebra approach, as in [Dekle, Eaton, and Kortum \(2007\)](#) and [Caliendo and Parro \(2015\)](#). Crucially, trade-side fundamentals, including productivities, trade costs, and input-output coefficients, either drop out of the solution completely or can be read directly off of the World Input Output Tables (WIOT) ([Timmer et al. 2015](#)).

4.2 Recovering Switching Costs

We recover switching costs C_{ij}^{ka} by inverting the model’s choice probabilities. In principle, data on worker flows across sectors would allow us to back out a full set of costs without parametric assumptions. In practice, the resulting matrix of transitions across sectors contains many zeros, and [Dingel and Tintelnot \(2025\)](#) argue that directly using such sparse matrices may produce misleading counterfactual predictions. The censoring of small cells required by SIAB data protection regulations

Table 6: Externally Calibrated Parameters

Parameter	Description	Value	Source
θ_i	Trade elasticities	6.638 ^a	Caliendo and Parro (2015)
β	Subjective discount factor	0.977 ^b	Annual discount factor = 0.96
λ^L	Low skill share	0.150	SIAB: share with no degree
λ^M	Medium skill share	0.730	SIAB: share with vocational degree
λ^H	High skill share	0.120	SIAB: share with university degree
χ	Young-old transition rate	0.192	Matches share of young workers
ζ	Retirement rate	0.449	(70%) and working lifespan (47 years)

Note: ^a We take estimates of sector-specific trade elasticities from Table 1 of [Caliendo and Parro \(2015\)](#). We enforce a lower bound of 3 and an upper bound of 12 on these elasticities. The table reports a simple average. ^b Incorporating the 0.37 probability of exit for old workers results in a population-average discount factor of 0.87 when $\beta = 0.977$, which, annualized, is approximately 0.97.

further compounds the problem.

We address these challenges in two steps, described in further detail in Supplemental Appendix C.1. First, we parameterize switching costs as a function of origin-sector and destination-sector fixed effects, the former also interacted with a dummy for same-sector transitions; a measure of skill distance between sectors; and bilateral dummies for transitions between manufacturing and non-manufacturing. Second, we run pseudo-Poisson maximum likelihood (PPML) regressions in the data that parallel this specification and solve for cost parameters such that the model exactly matches the estimated regression coefficients. This specification successfully matches empirical mobility patterns: the correlation between fitted and actual transition probabilities exceeds 0.99. The recovered costs C_{ij}^{ka} depend on the candidate Θ , because the mapping from observed flows to implied costs runs through the model's choice probabilities. Therefore, we calibrate switching costs jointly with Θ and henceforth write $C(\Theta)$.

4.3 The China Shock

Our calibration strategy requires a model analogue of the China shock. We define the shock in manufacturing sector i as the proportional change in Chinese productivity in that sector between 2000 and 2010. Following [Caliendo, Dvorkin, and Parro \(2019\)](#), we choose the vector of shocks such that the model-implied changes in Germany's net imports from China in each sector exactly match those predicted by our first-stage regressions in Section 2. Because the model solution depends on Θ , the implied shock vector $CS(\Theta)$ also varies with the candidate parameters.¹⁶

4.4 Calibrating Θ

We calibrate the 20 parameters in Θ by targeting 23 moments: 12 worker type and employment status-specific retraining rates, 5 empirical estimates of China shock effects from Table 4 as well as Column (2) of Table 5, and the coefficients from 6 type-specific versions of the retraining-switching regressions in Table 2. Formally, all the parameters are jointly identified by all the targeted moments. Nonetheless, we provide a heuristic identification argument in three steps.

Step 1: The type- and employment-status-specific retraining rates are closely tied to the retraining cost shifters F_0^{ka} and F_1^{ka} . These 12 parameters are effectively pinned down by the corresponding retraining rates. Thus, we set them aside and focus on the remaining 8 parameters – the sector preference dispersion ν , the retraining preference dispersion σ , and the 6 type-specific values of the retraining-switching complementarity Γ^{ka} – which target 11 moments.

Step 2: Four of the remaining target moments involve the labor market effects of the China shock, including the estimated coefficients for the earnings, employment, manufacturing employment, and sector switching outcomes from Section 2. These coefficients capture the responsiveness of the labor market to trade shocks and therefore vary directly with ν , the dispersion of sector preference

¹⁶In practice the China shock we infer is almost invariant to Θ .

shocks. With ν effectively pinned down by these four moments, the identification problem reduces to 7 parameters, i.e. σ and the 6 type-specific Γ^{ka} , targeting 7 moments.

Step 3: The first moment which helps identify sector preference dispersion and the retraining complementarities is the effect of the China shock on retraining rates, which we estimated in Column (2) of Table 5. However, this moment alone cannot separately identify σ and Γ^{ka} , as the expression for retraining rates in Theorem 1 clarifies:

$$\rho_i = \bar{\rho} + \bar{\rho}(1 - \bar{\rho}) \left(\frac{\Gamma}{\sigma} \right) (1 - \bar{\mu}_{ii}).$$

Consider a sector i highly exposed to the China shock.¹⁷ Then our empirical results in Section 2 imply that workers in this sector will switch to new sectors at high rates, which corresponds to a high $(1 - \bar{\mu}_{ii})$, and retrain at high rates, which corresponds to a high ρ_i . Given the expression above, we must then have $\Gamma > 0$, so that retraining complements switching. Crucially, however, the magnitude of the response of retraining to the China shock depends on both Γ and σ , so this moment alone cannot separately identify these two parameters.

To solve this problem, we additionally target 6 coefficients from 6 type-specific regressions of switching on retraining, equivalent to the specifications in Column (3) of Table 2 and reported in full in Supplemental Appendix Table S2. As in Table 2, we control for a rich set of observables; even so, we expect these regressions to confound the causal effect of retraining with selection on workers' unobserved propensity to switch sectors. For example, life events such as injuries may induce both retraining and switching. However, our model of the decision to retrain explicitly captures this selection problem. Because they make the two decisions simultaneously, workers in the model who draw a strong preference for switching sectors will be more likely to retrain, provided $\Gamma > 0$, and more likely to switch sectors even without retraining. By matching these regression coefficients to their model counterparts, we thus effectively control for selection on this unobserved propensity to switch sectors and thus discipline the complementarity between retraining and switching.

Algorithm

We calibrate Θ with a fixed point iteration. First, we start with a candidate Θ_0 and infer switching costs $C(\Theta_0)$ and the China shock $CS(\Theta_0)$. Second, we feed the China shock into the model and solve for the resulting steady state and transition path, yielding wages, transition probabilities, and retraining rates by worker type, sector, and period. Our solution takes the German and world economies in 2000 as the baseline equilibrium. Furthermore, we assume that the China shock is unforeseen in 2000, but workers learn of it between 2000 and the start of the next model period in 2005 — and thus take it into account when making their sector choice and retraining decisions for 2005. Third, using this simulated data, we run five regressions that parallel those in Table 4 and Column (2) of Table 5. Additionally, for each worker type, we compute average retraining rates and run switching-retraining regressions that parallel those in Supplemental Appendix Table

¹⁷Note that, in the simple model underlying Theorem 1, workers are homogeneous, so Γ has no type superscript.

S2.¹⁸ We thereby obtain a vector of model-implied moments $m(\Theta_0|C(\Theta_0), CS(\Theta_0))$, with a vector of standard errors denoted by se , and choose an updated value of Θ to solve

$$\Theta_1 = \underset{\Theta}{\operatorname{argmin}} \left\{ \sum_i \left(\frac{m_i - m_i(\Theta|C(\Theta_0), CS(\Theta_0))}{se_i} \right)^2 \right\}. \quad (30)$$

Finally, we repeat the steps above, replacing Θ_0 with Θ_1 , until the two converge.¹⁹

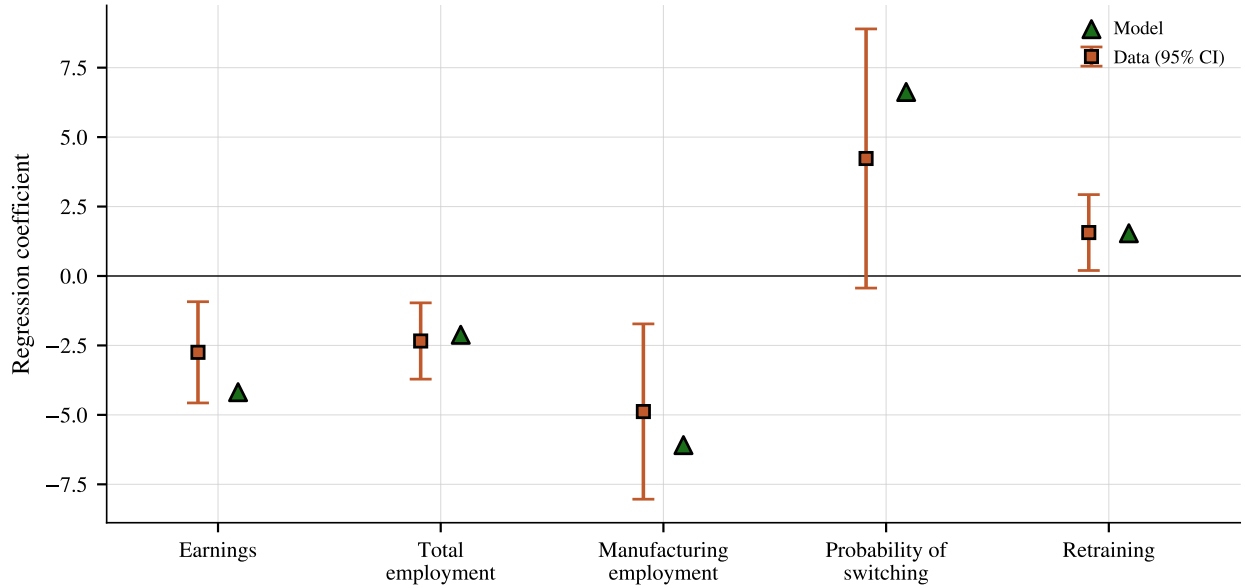
4.5 Calibration Results

Figure 2 shows that the model does a good job of matching the effects of the China shock on labor market outcomes, including retraining rates.²⁰ Table 7 reports the internally-calibrated parameter values as well as the mean switching costs for each worker type. We relegate the details of the China shock implied by our calibration to Supplemental Appendix C.4.

Dispersion Parameters

The first two rows of Panel A of Table 7 show our estimates of the sector and retraining preference dispersion parameters ν and σ , common to all worker types. The dispersion of sector preference

Figure 2: China shock effects in model and data



Note: Squares show empirical estimates from Tables 4 and 5 along with 95% confidence intervals. Triangles show model-implied coefficients at the parameter values in Table 7.

¹⁸We measure all model moments over two periods following the shock (i.e., 2005 and 2010), paralleling the empirical window.

¹⁹In practice, the dependence of the model-implied m on Θ via switching costs and the China shock is negligible, and this procedure converges on the second iteration.

²⁰The model fits the other 18 target moments perfectly. See Supplemental Appendix Table S6 for details.

shocks, $\nu = 0.307$, implies that a transitory 1% increase in real wages in a sector raises the probability that workers switch to that sector by roughly $\frac{\beta}{\nu} \simeq 3.18\%$. In Supplemental Appendix C.5, we show that this estimate lies around the middle of existing estimates in the literature, adjusting for differences in time horizon across papers. The dispersion of retraining shocks, $\sigma = 0.278$, implies that the retraining decision is quite sensitive to costs: a 10% increase in costs F results in a decline in retraining rates of between 0.67 and 2.21 percentage points, depending on a worker's age, skill, and sector of employment.

Switching Costs

The first row of Panel B, in turn, shows mean switching costs by worker type, averaged across origin-destination pairs and weighted by transition frequency. The mean across types, at 1.2, implies switching costs equivalent to a 70% reduction in real wages over five years — an economically large effect, but one nonetheless in line with existing estimates of sectoral transition costs (Artuç, Chaudhuri, and McLaren 2010; Dix-Carneiro 2014). Switching costs are fairly homogeneous across skill groups but 12-25% higher for old workers.

Costs and Benefits of Retraining

The final three rows of Panel B report the costs and benefits of retraining. We estimate substantial retraining costs, which average roughly 64% of the aforementioned switching costs for employed workers. As with switching, old workers face higher retraining costs. We also find that retraining is less costly for workers in non-employment, consistent with a potentially lower opportunity cost of time.

Table 7: Internally Calibrated Parameters

<i>Panel A: Common parameters</i>								
Parameter								Value
ν	Sector pref. dispersion							0.307
σ	Retraining pref. dispersion							0.278
<i>Panel B: Type-specific parameters</i>								
Parameter		(Y,L)	(Y,M)	(Y,H)	(O,L)	(O,M)	(O,H)	Avg.
$\bar{C}^{ka}$	Switching cost	1.090	1.206	1.078	1.362	1.355	1.291	1.206
F_0^{ka}	Retraining cost (non-emp.)	0.439	0.527	0.604	0.603	0.559	0.602	0.530
F_1^{ka}	Retraining cost (emp.)	0.481	0.766	0.911	0.968	0.922	1.034	0.774
Γ^{ka}	Complementarity	0.357	0.367	0.222	0.356	0.396	0.405	0.357

Note: Worker types are indexed by age (Y = young, O = old) and skill (L = low, M = medium, H = high). Table reports the parameter values that minimise the loss function defined in (30), as well as the switching costs we infer from observed worker flows at these parameter values. Switching costs in the table are an average over $i - j$ transitions, weighted by the baseline frequency of those transitions.

Furthermore, we estimate a clear complementarity between retraining and switching. Our calibration finds positive values for the retraining complementarity parameter Γ^{ka} across all worker types — hence, retraining uniformly allows workers to more easily switch sectors. Averaging across all types, the calibrated values of Γ^{ka} in the final row of Panel B imply that retraining lowers the effective cost of switching by a full 30%. This retraining-switching complementarity remains fairly homogeneous across worker types, with the notable exception of young, high-skilled workers, who seem to benefit less.

As a result of the high switching costs combined with a strong retraining-switching complementarity, workers who retrain are much more likely to move to a new sector. Averaging over all workers, retraining approximately *doubles* the probability of switching sectors, to 0.38 from a baseline of 0.18. Of course, the subset of workers who actually retrain are positively selected on their idiosyncratic propensity to switch, but even within this group retraining has large effects: their probability of moving to a new sector rises from 0.42 to 0.66 as a result of retraining.

In Supplemental Appendix C.2, we decompose the welfare gain from retraining for an individual worker into three components: the direct gain, or idiosyncratic benefit minus cost; the intensive margin gain from sector switching, i.e. the reduction in switching costs for workers who would switch sectors regardless of whether they retrain; and the extensive margin gain from switching to a new sector for workers who only switch when they retrain. Three findings emerge from this decomposition. First, workers who retrain experience welfare gains equivalent to a 31% increase in real wages over five years. Second, direct gains are modest (11%) – in other words, retrained workers benefit mainly from the complementarity between retraining and switching. Third, intensive margin gains (15%) dominate extensive margin gains (5%) – most of the welfare benefit comes from easing transitions that workers would have made anyway.

5 Quantifying the Role of Retraining

With the calibrated model in hand, we now turn to our paper’s central question: how does retraining shape the effects of trade shocks? We first study the China shock in detail: we contrast its effects in the baseline economy against its effects under different retraining scenarios. We then conclude with a briefer analysis of the effects of Donald Trump’s “Liberation Day” tariffs on Germany.

5.1 The China Shock in the Baseline Economy

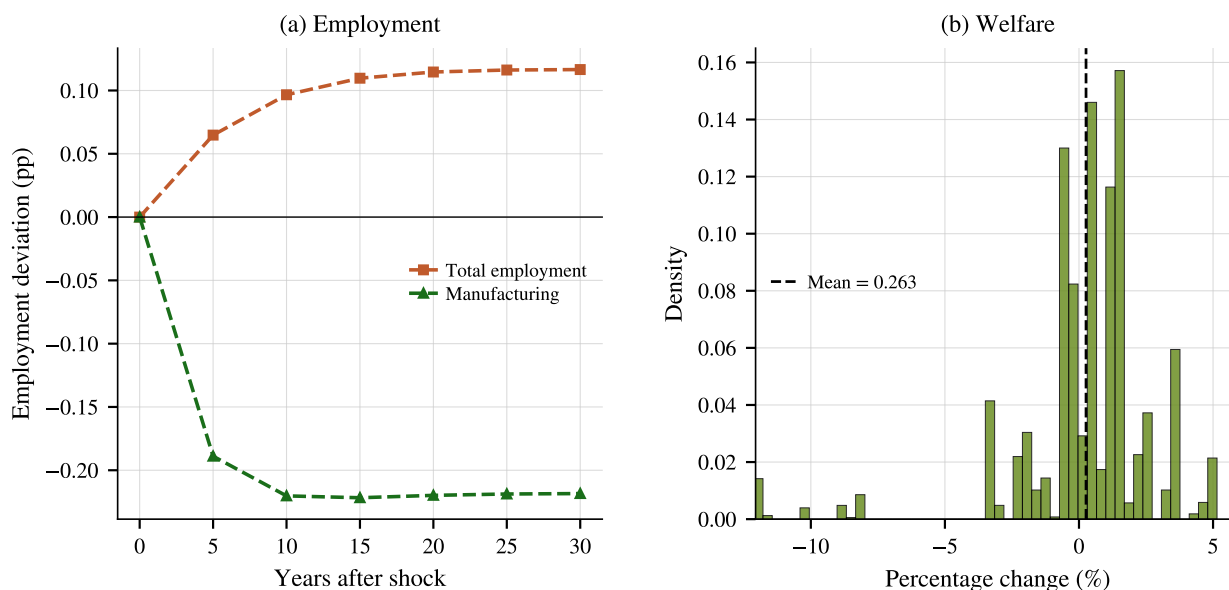
We start with the baseline economy — with retraining costs as calibrated in Table 7 — and simulate our calibrated “China shock,” or changes in manufacturing productivities in China. We model this shock exactly as during calibration: the period zero equilibrium, which need not be a steady state, corresponds to the world economy in 2000. Between period zero and period one, German workers learn about the China shock and incorporate it into their retraining and sector-switching decisions. The productivity shock arrives in period one and is permanent.

Aggregate employment shares move relatively little. Panel (a) of Figure 3 shows that the overall employment rate rises by 0.12 percentage points, as cheaper imported goods draw workers out of non-employment, while manufacturing employment declines by 0.22 percentage points.

These small aggregate shifts mask substantial heterogeneity across manufacturing sectors. Employment in textiles collapses (−44%) and employment in computers and electronics also falls sharply (−19%), while chemicals grows by 7% and motor vehicles by 6% — consistent with China’s growing comparative advantage in labor-intensive manufacturing in this period.²¹

How did this shock affect the welfare of German workers? As in Section 3, we measure changes in welfare using equivalent variation (*EV*) expressed as a fraction of consumption in a single period.²² The China shock allows German consumers and producers to benefit from cheaper imports, and ultimately raises baseline-employment-weighted average welfare by 0.45%. The histogram in Panel (b) of Figure 3 focuses on workers initially employed in manufacturing and shows the employment-weighted distribution of *EV* within this group. The average manufacturing worker experiences positive, although smaller, welfare gains (0.26%). More strikingly, welfare effects within

Figure 3: Baseline effects of the China Shock



Note: Panel (a) shows percentage point changes in total employment (circles) and manufacturing employment (triangles). The difference is taken between the economy with the baseline retraining parameters in Table 7 and the China shock, and the economy with the same parameters but no shock. Panel (b) shows the equivalent variation of the China shock, expressed as a percentage of income in a single period, among manufacturing workers. Equivalent variation is defined for each age-skill-initial employment cell, and cells are weighted by their baseline employment. The vertical line shows the baseline employment-weighted mean equivalent variation.

²¹The decline in employment in textiles implied by the model is so large that we might expect to see it in the raw data: and indeed, between 1999 and 2008 employment in NACE 15 (manufacture of textiles) and NACE 18 (manufacture of apparel) fell by 33% and 44% respectively (Görg and Görllich 2011).

²²Caliendo, Dvorkin, and Parro (2019) use the same welfare definition but express *EV* as a fraction of consumption in every period. Discounting *EV* in this way is not straightforward in our model because young and old workers have different discount rates.

this group are widely dispersed. The China shock makes a full 37.0% of manufacturing workers worse off, and the left tail of the distribution is long — the welfare of the worst-affected worker type and initial sector falls by 12.13%.

Workers' initial sector alone accounts for just over 90% of the total variation in EV . Workers initially in textiles lose 10.3%, on average, and those in computers and electrical equipment, 3.0%, while workers in basic metals gain 4.2% and those in motor vehicles, 2.9%. Age and skill matter too, but far less than initial sector, as Supplemental Appendix Table S9 documents. Young workers gain more on average than old workers (0.591% vs. 0.114%), and 40% of low-skill manufacturing workers are made worse off by the shock, compared to 37% of medium-skill and 33% of high-skill workers.

Overall, the effects of the China shock on Germany qualitatively mirror the impacts [Caliendo, Dvorkin, and Parro \(2019\)](#) find on the United States, in spite of the vast structural differences between the two economies. In particular, they find that aggregate employment rises by 0.22 percentage points, that manufacturing employment falls by 0.36 percentage points, and that the shock has positive welfare effects, on average. As in our German case, they find substantial inequality across sectors — as well as, in their case, local labor markets.

5.2 Retraining and the China Shock

How does retraining interact with the China shock? Starting from the baseline economy, we now simulate two shocks simultaneously: the China shock studied above and a multiplicative change Δ in retraining costs, applied equally to workers of every age, skill, and employment status. We then track employment, earnings, and welfare as the economy converges towards a new steady state. To isolate the effect of the China shock at a given time t , we compare the path of a variable under both shocks to its path with the retraining cost change alone. That is, we difference out the direct effect of the retraining policy and focus on how the presence of retraining alters the impact of the China shock.

We consider four scenarios for retraining costs: $\Delta = \infty$ ("No retraining"); $\Delta = 1$ ("Baseline"); $\Delta = 0.65$, which causes the retraining rate to roughly double ("2x"); and $\Delta = 0.45$, which causes the retraining rate to roughly triple ("3x"). Table 8 reports our main results. The first row shows the counterfactual retraining rate and the second shows the resulting aggregate rate at which workers switch sectors. Consistent with Theorem 1, as the retraining rate rises, so does the aggregate sector switching rate: from 28.2% in the *No retraining* scenario to 36.1% in the 3x scenario. The rest of the table explores how these changes in mobility alter the effects of the China shock. We focus on welfare and employment dynamics, as well as the role of general equilibrium forces and heterogeneity across workers.

Welfare

Germany's aggregate welfare gains from the China shock are nearly invariant to the retraining rate, as shown in the first panel of Table 8. Thus, the insight from Theorem 2 that retraining does not affect aggregate welfare in the simple model generalizes to this richer setting. The small differences in aggregate welfare effects across retraining scenarios reflect compositional effects absent from the simple model: the China shock reduces non-employment, and non-employed workers benefit more from retraining than employed workers. As a result, the China shock slightly lowers the aggregate gains from retraining, such that the welfare gains from the China shock fall as retraining costs decrease and retraining rates rise. Among workers initially in manufacturing, the sign of this relationship is reversed, but again the magnitudes are tiny.

Retraining does, however, substantially compress the distribution of the welfare effects of the China shock, as laid out in the second panel of Table 8. Here again, the Theorem 2 intuition regarding the option value of mobility generalizes: retraining lowers the cost of switching sectors and so compensates workers in adversely affected sectors with the option of moving to a new one. From the *No retraining* column to the 3x column, the standard deviation of welfare changes across all

Table 8: Effects of the China Shock Across Retraining Scenarios

	No retraining	Baseline	2x	3x
Retraining rate	0.0%	11.5%	22.9%	34.4%
Switching rate	28.2%	31.1%	33.8%	36.1%
<i>Welfare: aggregate</i>				
Mean EV	0.47	0.45	0.44	0.43
Mean EV (mfg)	0.26	0.26	0.27	0.28
<i>Welfare: distribution</i>				
Std(EV)	1.36	1.24	1.13	1.05
Std(EV) (mfg)	2.90	2.63	2.41	2.22
% of mfg. workers with EV < 0	37.5	37.0	37.0	37.0
Min EV	-13.60	-12.13	-10.88	-10.12
<i>Aggregate employment dynamics</i>				
Aggregate employment (pp change), year 5	0.06	0.07	0.07	0.06
Aggregate employment (pp change), LR	0.12	0.12	0.11	0.10
Mfg. employment (pp change), year 5	-0.19	-0.19	-0.19	-0.20
Mfg. employment (pp change), LR	-0.22	-0.22	-0.22	-0.22
<i>Sectoral employment dynamics (mfg)</i>				
Short-run reallocation	4.84	4.90	4.97	5.04
Long-run reallocation	7.23	7.10	7.01	6.93
% of LR. at year 5	66.8	68.9	70.9	72.6

Note: Each column shows effects of the China shock under different retraining cost scenarios. The effects of the China shock are always defined as differences between the economy with and without this shock, so that direct effects of changes in retraining costs are differenced out. Means and standard deviations are weighted by baseline employment. "Mfg" refers to manufacturing. Our measure of reallocation at year t is the average of $|l_{mt} - l_{m0}|$, where m indexes manufacturing sectors and employment l_{mt} is measured as a percentage of total manufacturing employment in period t . Short run refers to $t = 1$ and long run to $t \rightarrow \infty$.

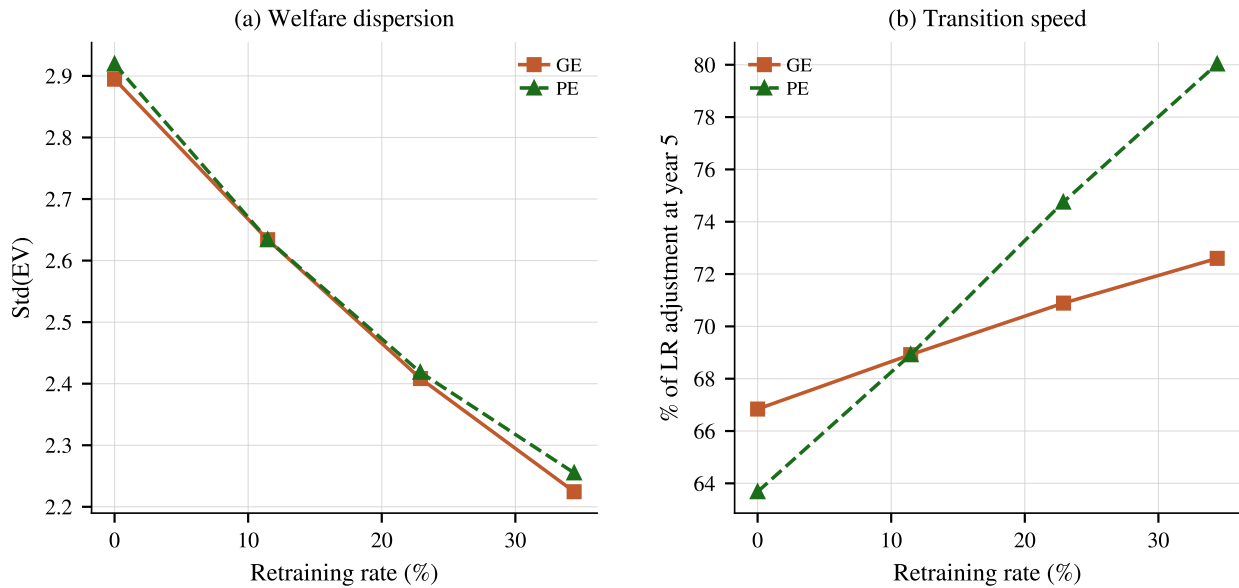
workers falls by 22.9% and across manufacturing workers by 23.2%. Panel (a) of Figure 4 illustrates this decline. At the bottom end of the distribution, the share of manufacturing workers who lose from the shock falls marginally, from 37.5% to 37.0%, while the welfare loss of the worst affected workers — old, low-skill workers initially employed in textiles — declines more substantially, from -13.60% to -10.12% .

We thus conclude that retraining does relatively little to shrink the set of adversely-affected workers but does limit the size of this group's losses. Expanding retraining programs can blunt the distributional conflicts that concern policymakers, but the magnitudes in Table 8 do suggest that, even in an economy with widely available retraining, some workers will inevitably lose from the expansion of international trade.

Employment Dynamics

In the third panel of Table 8, we show that retraining has small, non-monotone effects on aggregate and manufacturing employment changes in both the short and the long run. To focus on employment dynamics within manufacturing, we measure short-run reallocation as the average absolute percentage point change in employment shares across manufacturing sectors at $t = 1$, i.e. at year five, and long-run reallocation as the same quantity as $t \rightarrow \infty$. In the final row we also report a

Figure 4: Welfare and adjustment under varying retraining costs



Note: Both panels show how an outcome varies across retraining cost scenarios. In both panels the solid line (“GE”) shows the full model and the dashed line (“PE”) shows the partial equilibrium model in which the path of real wages in every sector is held constant even as retraining costs vary. Panel (a) shows the baseline employment-weighted standard deviation of equivalent variation across workers initially in manufacturing. Panel (b) shows our measure of transition speed, the fraction of long-run employment reallocation within manufacturing that is complete within five years of the shock.

natural measure of transition speed: the share of the long-run reallocation complete at $t = 1$.²³ Under the *Baseline* scenario, the average manufacturing sector sees a 4.90 percentage point change in its employment share in the short run, which grows to 7.10 percentage points in the long run; 68.9% of long-run reallocation occurs within five years.

Looking across columns, the reduction in effective switching costs caused by retraining makes it easier for workers to exit declining sectors and enter booming ones, resulting in larger employment share changes in the *short run*. But in the *long run*, in line with Theorem 3, the pattern reverses: higher retraining rates go hand-in-hand with more limited sectoral reallocation. Intuitively, with lower retraining costs and thus higher retraining rates, the option value of switching in the future better compensates workers in low-wage sectors, so their sectoral choices are less responsive to wages. This combination of larger short-run and smaller long-run responses under lower-cost retraining then serves to increase the speed of the transition to the new steady state. The share of the long-run transition complete within five years rises from 66.8% in the *No retraining* column to 72.6% in the *3x* column, as also plotted in the solid line in Panel (b) of Figure 4.

Heterogeneity

Retraining has small effects on the average gains or losses of each worker type but important distributional effects within each type. Supplemental Appendix Table S9 disaggregates the welfare effects of retraining by worker age and skill. Retraining has larger effects on young workers: the standard deviation of *EV* among young manufacturing workers falls by 25% as we move from the *No retraining* scenario to the *3x* scenario, but by only 15% for old workers. This difference is driven by lower baseline retraining rates among old workers, as well as lower baseline mobility across sectors. Across skill groups, retraining has similar effects on low and medium skill workers but matters much less for high skill workers, whose baseline retraining rate is just 6%.

Partial Versus General Equilibrium

Retraining not only directly affects switching costs but may also have general equilibrium effects on wages through its impact on employment reallocation. To isolate this channel, Supplemental Appendix Table S8 repeats the retraining-cost experiments with real wages held fixed along their trajectory under the *Baseline* scenario in Table 8 — that is, under the China shock with retraining costs at their calibrated values.

The dashed lines in each panel of Figure 4 show how the dispersion of welfare effects and the speed of the transition vary with retraining rates under this partial equilibrium counterfactual. The dotted partial-equilibrium line in Panel (a) almost perfectly overlaps with the solid general-equilibrium line: general equilibrium effects hardly matter for the relationship between retraining and the dispersion of welfare. But the two lines in Panel (b) diverge sharply; evidently, general equilibrium forces do matter for how retraining shapes employment dynamics. Retraining shifts

²³This can also be written as weighted average of the sector-level convergence speeds defined in Theorem 3.

workers out of some sectors and into others; in general equilibrium these shifts are partially offset by countervailing moves in real wages, but the partial equilibrium model ignores this feedback and so overstates how much retraining can speed up labor reallocation. In the full general equilibrium model, the share of the transition completed within five years rises from 66.8% under *No retraining* to 72.6% in the 3x scenario; in contrast, the dashed line in Panel (b) shows that in partial equilibrium it rises from 63.7% to 80.0% — about three times as large an effect.

5.3 US Trade Policy Shock

To understand how far these quantitative results generalize, we now consider a second counterfactual shock. Loosely inspired by Donald Trump’s “Liberation Day” tariff announcement, we feed the model a 20% increase in Germany’s iceberg costs of exporting to the United States.²⁴ For comparability with our China shock exercise above, we start this counterfactual experiment from the same baseline economy, make the same timing assumptions, and consider the same range of retraining cost scenarios. Supplemental Appendix Table S10 details the results.

In the economy with baseline retraining costs, this “Trump tariff shock” lowers aggregate employment by 0.22 percentage points, manufacturing employment by 0.09 percentage points, and aggregate welfare by 0.85%. As in the case of the China shock, the level of retraining costs primarily affects the dispersion of *EV* and the speed of transition. Going from *No retraining* to 3x, the standard deviation of *EV* across all workers falls modestly (8.3%), and by slightly more (12.6%) across manufacturing workers. Over the same range, the fraction of labor market reallocation completed within five years rises from 63.2% to 68.7%.

While the effect of retraining on transition speed mirrors the China shock case, its effects on welfare are roughly a third as large across all workers, and half as large in manufacturing. The explanation lies in the much more uniform effects of the Trump tariff shock across sectors. The standard deviation of *EV* across workers is just 0.46%, and the shock leaves essentially all manufacturing workers worse off. Crucially, even this limited inequality does not align with workers’ initial sectors: between-sector variation in *EV* accounts for about 25% of the total, versus 90% for the China shock. Since moving sectors does little to help a worker escape the adverse effects of such a shock, retraining — a technology which eases mobility across sectors — is less effective.

5.4 Taking Stock

Our counterfactual experiments deliver three main results. First, retraining mitigates inequality in the welfare effects of trade shocks: in an economy with a retraining rate triple the status quo, the dispersion of the China shock’s welfare effects falls by roughly one-fifth, relative to an economy in which retraining is impossible. Second, retraining helps the economy reach its new steady state

²⁴A more complete exercise would account for the fact that tariff revenue is rebated to US consumers and so has different general equilibrium effects. The feedback to Germany of such a rebate is likely to be negligible.

employment distribution more quickly, although general equilibrium feedback to wages sharply limits the magnitude of this force. Third, retraining proves more useful for trade shocks with unequal effects across sectors, e.g. the China shock, than for more uniform trade policy shocks.

6 Conclusion

In this paper, we study the ability of government-subsidized retraining to help workers adjust to international trade shocks. Using administrative data from Germany, we find that retraining is tightly linked to transitions across sectors and that workers retrain in response to rising import competition from China. We then develop a model in which workers jointly decide whether to retrain and whether to switch sectors and show that, when the two are complements, increases in the availability of retraining effectively reduce sector-switching costs. Our calibrated parameters reveal a complementarity between retraining and sector switching for all worker skill levels and ages: the average worker who retraines faces 30% lower effective switching costs, relative to her unretrained counterparts. In our counterfactuals, lower retraining costs go hand-in-hand with greater short-run and more limited long-run sectoral employment reallocation. In consequence, retraining dampens inequality in the effects of trade but scarcely alters its aggregate welfare impacts.

Our results highlight retraining as a useful policy tool for managing the distributional effects of sectoral shocks: it evens out the welfare effects of import competition and accelerates labor market adjustment without diminishing the aggregate gains from trade. Even so, large sectoral shocks still produce losers, so retraining should complement rather than replace other adjustment policies.

Appendix

A Equivalent Variation Derivation

In the context of the simple model of Section 3, we define the *equivalent variation* of a shock to world prices for workers initially in sector i as the proportional transfer that would leave them indifferent between receiving this transfer versus experiencing the shock. As in Section 3, we assume that workers learn about this shock between $t = 0$ and $t = 1$. Denote worker values with the transfer but without the shock to world prices with a 0 superscript, and denote values without the transfer and with the shock with a 1 superscript. Then at $t = 0$:

$$v_{i0}^0 = \log(w_{i0}(1 + EV_i)) + U_{i0}^0, \quad v_{i0}^1 = \log(w_{i0}) + U_{i0}^1,$$

noting that real wages at $t = 0$ are unaffected by the shock. Differencing these equations and using the fact that $v_{i0}^0 = v_{i0}^1$ by definition immediately gives,

$$\log(1 + EV_i) = U_{i1}^1 - U_{i0}^0. \quad (\text{A.1})$$

Assuming that the shock is small, and thus that $\log(1 + EV_i) \simeq EV_i$, gives the definition used in Section 3. Whenever we conduct quantitative counterfactuals in Section 5, we directly use the formula (A.1) for each worker type ka . Notice that in the quantitative model we implicitly assume that all OLG transitions (young workers ageing and old workers retiring) occur before workers learn about the shock.

B Proofs of Analytical Results

Proof of Theorem 1. We proceed in three steps: solve the model at $\Gamma = 0$, differentiate with respect to Γ , and evaluate the derivative at $\Gamma = 0$.

Step 1: Solution at $\Gamma = 0$. When $\Gamma = 0$ the retraining decision does not depend on the sector choice. A worker retrain if and only if $\sigma\eta - F > 0$, giving a retraining rate $\bar{\rho} = \Pr(\eta > F/\sigma) = (1 + e^{F/\sigma})^{-1}$, which is common across sectors. Since the retraining payoff $\max\{\sigma\eta - F, 0\}$ is the same regardless of destination, it cancels from pairwise comparisons. The remaining sector choice is a standard logit, giving

$$\log\left(\frac{\bar{\mu}_{ij}}{\bar{\mu}_{ii}}\right) = \frac{\beta(v_j - v_i) - C}{v}.$$

Step 2: Differentiation. We derive $\frac{\partial}{\partial \Gamma} \log(\mu_{ij}/\mu_{ii})|_{\Gamma=0}$. Conditional on the retraining shock η , the sector choice remains logit, with the payoff of each switching option $j \neq i$ augmented by $\max\{\Gamma + \sigma\eta - F, 0\}$ and the payoff of staying augmented by $\max\{\sigma\eta - F, 0\}$. Differentiating with respect to Γ : the payoff of each switching option $j \neq i$ changes at rate $\mathbb{1}\{\Gamma + \sigma\eta - F > 0\}$, which at $\Gamma = 0$

equals $\mathbb{1}\{\eta > F/\sigma\}$, while the payoff of staying in i does not depend on Γ . Since the payoffs of all switching options change by the same amount, the standard logit derivative gives

$$\left. \frac{\partial \mu_{ij}(\eta)}{\partial \Gamma} \right|_{\Gamma=0} = \frac{\mathbb{1}\{\eta > F/\sigma\}}{\nu} \bar{\mu}_{ij} \bar{\mu}_{ii}, \quad \left. \frac{\partial \mu_{ii}(\eta)}{\partial \Gamma} \right|_{\Gamma=0} = -\frac{\mathbb{1}\{\eta > F/\sigma\}}{\nu} \bar{\mu}_{ii} (1 - \bar{\mu}_{ii}).$$

At $\Gamma = 0$ the conditional probabilities $\mu_{ij}(\eta) = \bar{\mu}_{ij}$ do not depend on η , so the unconditional derivatives are obtained by taking expectations over η , replacing $\mathbb{E}[\mathbb{1}\{\eta > F/\sigma\}]$ with $\bar{\rho}$. Therefore

$$\left. \frac{\partial}{\partial \Gamma} \log \frac{\mu_{ij}}{\mu_{ii}} \right|_{\Gamma=0} = \frac{1}{\bar{\mu}_{ij}} \frac{\partial \mu_{ij}}{\partial \Gamma} - \frac{1}{\bar{\mu}_{ii}} \frac{\partial \mu_{ii}}{\partial \Gamma} = \frac{\bar{\rho} \bar{\mu}_{ii}}{\nu} + \frac{\bar{\rho}(1 - \bar{\mu}_{ii})}{\nu} = \frac{\bar{\rho}}{\nu}.$$

A first-order expansion around $\Gamma = 0$ then yields (6). Expression (8) follows from evaluating the logit choice probabilities at the effective switching cost $C - \Gamma \bar{\rho}$.

Step 3: Retraining rate. Decompose the retraining probability as

$$\rho_i(\Gamma) = \mathbb{E}_\eta[\mathbb{1}\{\eta > F/\sigma\} \mu_{ii}(\eta, \Gamma)] + \mathbb{E}_\eta[\mathbb{1}\{\eta > (F - \Gamma)/\sigma\} (1 - \mu_{ii}(\eta, \Gamma))].$$

Differentiating this with respect to Γ and evaluating at $\Gamma = 0$ gives

$$\left. \frac{\partial \rho_i}{\partial \Gamma} \right|_{\Gamma=0} = f(F/\sigma)(1 - \bar{\mu}_{ii}) = \bar{\rho}(1 - \bar{\rho})(1 - \bar{\mu}_{ii}),$$

where f is the logistic density. Equation (7) follows as a first-order approximation. $\square$

Proof of Theorem 2. *Part (i).* Equation (9) follows directly from $w_{it} = \tilde{w}_{it} P_t^{-1}$ with $\tilde{w}_{it} = P_{it}$ and the Cobb-Douglas price index $P_t = \prod_j P_{jt}^{\alpha_j}$.

Part (ii). Following the permanent shock, real wages, worker values, and choice probabilities are all constant over time. By the envelope theorem

$$EV_i = dU_i = \beta \sum_j \mu_{ij} dv_j = \beta \sum_j \mu_{ij} d \log w_j + \beta \sum_j \mu_{ij} dU_j = \beta \sum_j \mu_{ij} d \log w_j + \beta \sum_j \mu_{ij} EV_j.$$

Summing over i and using the fact that the initial allocation is a steady state yields $(1 - \beta) \overline{EV} = \beta \sum_i l_i d \log w_i$. Then substituting the expression for real wages in (i) gives the result.

Part (iii). Define $\tilde{m} = \left(\frac{N}{N-1}\right) (1 - \mu_{ii})$, noting that because the baseline is symmetric this quantity does not vary across sectors, and $\mu_{ij} = \tilde{m}/N$ for every $j \neq i$. Then the expression for EV_i in (ii) can be written

$$EV_i = \beta(1 - \tilde{m}) d \log w_i + \beta \tilde{m} \overline{d \log w} + \beta(1 - \tilde{m}) EV_i + \beta \tilde{m} \overline{EV},$$

where $\overline{d \log w} = \sum_j l_j d \log w_j$. Substituting $\overline{d \log w} = (1 - \beta) \overline{EV} / \beta$ from (ii) and rearranging, using the definition of λ in Theorem 2, gives the result. $\square$

Proof of Theorem 3. We maintain the first-order approximation of Theorem 1 and linearize around the symmetric steady-state baseline, at which $l_i = 1/N$, $\mu_{ij} = \tilde{m}/N$ for $j \neq i$, and $\mu_{ii} = 1 - (N - 1)\tilde{m}/N$, with $\tilde{m} \equiv \tilde{m}(\bar{\rho})$ as in the proof of Theorem 2. Under this approximation the sector choice probabilities are logit, so

$$d \log \mu_{ij} = \frac{\beta}{\nu} \left(dv_j - \sum_k \mu_{ik} dv_k \right) \quad (\text{A.2})$$

for any i, j . We log-linearize the law of motion $l_{it+1} = \sum_j \mu_{jit} l_{jt}$ and after some algebra obtain

$$d \log l_{it+1} = (1 - \tilde{m}) d \log l_{it} + \tilde{m}(2 - \tilde{m}) \left(\frac{\beta}{\nu} \right) (dv_i - dV),$$

where $dV = N^{-1} \sum_j dv_j$. Combining Part (iii) of Theorem 2 with the fact that $dv_i = d \log w_i + EV_i$ then yields,

$$d \log l_{it+1} = (1 - \tilde{m}) d \log l_{it} + \tilde{m}(2 - \tilde{m}) \left(\frac{1 - \beta\lambda}{\nu} \right) \left(\left(\frac{\beta}{1 - \beta} \right) d \log w_i - \bar{EV} \right).$$

Part (i) of Theorem 3 follows by setting $t = 0$ and noting that $d \log l_{i0} = 0$ for all i ; (ii) follows by setting $d \log l_{it+1} = d \log l_{it} = d \log l_{i\infty}$; and (iii) follows by rewriting the difference equation above as

$$\frac{d \log l_{it+1}}{d \log l_{i\infty}} = (1 - \tilde{m}) \frac{d \log l_{it}}{d \log l_{i\infty}} + \tilde{m} = 1 - (1 - \tilde{m})^{t+1}.$$

Comparative statics. The long-run coefficient $(2 - \tilde{m})(1 - \beta\lambda)/\nu$ is decreasing in $\tilde{m}$: both factors fall, since λ is increasing in $\tilde{m}$. This establishes part (ii). The short-run coefficient is $\tilde{m}$ times the long-run coefficient, equal to $\tilde{m}(2 - \tilde{m})(1 - \beta\lambda)/\nu$. The factor $\tilde{m}(2 - \tilde{m})$ is increasing in $\tilde{m}$ for $\tilde{m} \in (0, 1)$ while $(1 - \beta\lambda)$ is decreasing; for $\tilde{m}$ small the first effect dominates, establishing part (i). $\square$

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Supplemental Appendix

A Data Appendix

A.1 Additional Dataset Details

A.1.1 German Sample of Integrated Labor Market Biographies (SIAB)

We measure labor market outcomes using the Sample of Integrated Labor Market Biographies (SIAB) 1975-2021 (Graf et al. 2023), accessed on-site and via remote access at the Research Data Centre (FDZ) of the Institute for Employment Research (IAB). The SIAB is a 2% sample from Germany’s Integrated Employment Biographies (IEB), recording, for each worker and month between 1975 and 2021, employment spells, unemployment benefits, and participation in active labor market programs such as retraining (Schmucker, Seth, and Berge 2023).

General Cleaning and Preparation. We clean the SIAB using the Stata routines of Dauth, Eppelsheimer, and Stüber (2023), which split episodes spanning multiple calendar years, account for lump-sum payments, calculate job tenure, merge in firm characteristics, aggregate education levels, merge in the contribution assessment ceiling and marginal part-time income threshold, and deflate monetary values. We restrict the panel to workers aged 18-65, drop employment spells with zero income, and drop employment statuses not observed in all years: marginal employment spells, internships, and a catchall “other” category. We then apply the Dauth, Eppelsheimer, and Stüber (2023) imputation procedure to correct for wage top-coding and their final cleaning routine.

Sector Classification. In our reduced form empirics, we use a slightly-aggregated version of ISIC Rev. 4 sectors, which we refer to as *reduced form sectors*. Each row of the first column of Table S1 lists the ISIC Rev. 4 sector code(s) corresponding to one of our reduced form sectors.²⁵ For our quantitative model calibration and counterfactual exercises, we use a slightly coarser classification, which we henceforth term *model sectors*, which combines several closely related sets of these reduced form sectors into larger groups. We report the complete crosswalk between our reduced form sectors, World Input-Output Database (WIOD) sectors, and our model sectors in Table S1. For robustness, we also re-estimate key empirical specifications at the level of the 3-digit sectors of the German Classification of Economic Activities, Edition 1993 (WZ 93).

Reduced Form Sample. Applying the Dauth, Eppelsheimer, and Stüber (2023) parallel-episode correction to the cleaned panel, we keep only a single “primary” job per worker, selected by longest firm tenure and breaking ties by (in order) daily earnings, episode length, and at random. We define a worker’s sector using our reduced form sector classification applied to this primary job

²⁵Note that ISIC Rev. 4 sector codes equal the first two digits of the German Classification of Economic Activities, Edition 2008 (WZ 08).

Table S1: Sector Classification

Reduced form sectors (ISIC Rev. 4 codes)	WIOD sector	Model sector	Description
<i>Panel A: Manufacturing</i>			
10	C10	6	Food
11	C11	6	Food
12	C12	6	Food
13	C13	7	Textiles
14	C14	7	Textiles
15	C15	7	Textiles
16	C16	8	Wood
17	C17	9	Paper
18	C18	10	Printing
19	C19	11	Coke
20	C20	12	Chemicals
21	C21	13	Pharma
22	C22	14	Rubber
23	C23	15	Minerals
24	C24	16	Basic metals
25	C25	17	Fabricated metals
26	C26	18	Computers
27	C27	19	Electrical
28	C28	20	Machinery
29	C29	21	Motor vehicles
30	C30	22	Other transport
31	C31	23	Other
32	C32	23	Other
33	C33	24	Repair
<i>Panel B: Non-manufacturing and non-employment</i>			
–	–	1	Non-employment
1–9	A01-A03, B	25	Non-manufacturing tradables
35–43	–	2	Utilities, construction
45–56	–	3	Trade, transport, hospitality
58–82	–	4	Communications, business services
84–99	–	5	Public administration, education, health

Notes: This table displays the sectoral classification, in the first column, used for our reduced form empirical analysis, listing the ISIC Rev. 4 codes corresponding to each sector, which also equal the first two digits of the German Classification of Economic Activities, Edition 2008 (WZ 08). The second column reports the corresponding WIOD sector codes. The third column displays the mapping to the model sectors used in the calibration and counterfactual exercises. The final column provides a brief description of each model sector.

and record the sum of earnings across all parallel jobs. Following [Dauth, Findeisen, and Suedekum \(2021\)](#), we take a cross-sectional snapshot for the year 2000 which includes all workers who, in their

primary job, were full-time employed in manufacturing on 30 June 2000, had at least two years of tenure, earned above the marginal earnings threshold, and were aged 24-65 throughout 2001-2010.²⁶ We use this *reduced form sample* for the sector switching on retraining regressions in Table 2 and Supplemental Appendix Table S2, the labor market outcome and retraining regressions in Tables 4 and 5 in the main text, as well as the retraining decomposition in Supplemental Appendix A.5. The labor market and retraining outcome regressions in Supplemental Appendix Table S3 use this same sample but instead measure workers' sectors and thereby trade exposure using the 3-digit sectors from the German Classification of Economic Activities, Edition 1993 (WZ 93).

Retraining Episode Sample. We define retraining as employment status (*erwstat*) codes 10001 (activation and vocational integration), 10002 (career choice and vocational training), and 10003 (vocational retraining and further education) in the Participants-In-Measures History File (MTH/XMTH) module of the SIAB. The SIAB only includes such episodes starting in 2000, so we restrict our *retraining episode sample* to spells between 2000 and 2019 and calculate episode length in days.

Quarterly Panel. Adapting Dauth, Eppelsheimer, and Stüber (2023)'s yearly-panel code, we build a *quarterly panel* of workers from 1990 through 2019 using the cleaned, parallel-episode-corrected SIAB. We again restrict the sample to workers aged 18-65 and measure employment status, earnings, sector, and primary-job characteristics on the midpoint date of each quarter. The quarterly retraining dummy equals one if the worker participates in subsidized retraining at any point during the quarter, not only on the sample date. Worker-quarters missing from the data are classified as non-employment. We additionally record non-employed workers' last observed sector, earnings, nationality, education, and job tenure.

A.1.2 UN Comtrade Database

Our trade data comes from the UN Comtrade Database.²⁷ Comtrade provides primary values by importing and exporting country and SITC Rev. 3 sector; we map these values to the 3-digit sectors in the German Classification of Economic Activities, Edition 1993 (WZ 93) with a crosswalk provided by Dauth, Findeisen, and Suedekum (2021). We then convert all values to 2005 USD using the FRED Consumer Price Index for All Urban Consumers: All Items in US City Average (CPIAUCSL), seasonally adjusted, indexed to January 2005 and convert to 2005 Euros using December 2005 exchange rates.²⁸ We then convert to 2015 Euros using the German CPI values from the German Statistical Agency as provided by Dauth, Eppelsheimer, and Stüber (2023). Finally, we crosswalk trade flows to a slightly-aggregated version of ISIC Rev. 4 sectors using establishment counts as provided for Germany by Eberle et al. (2014). We then focus on trade

²⁶Manufacturing sectors in the 3-digit German Classification of Economic Activities, Edition 1993 include sectors 151-212 and 241-366.

²⁷Available at <https://comtradeplus.un.org>.

²⁸CPI available at <https://fred.stlouisfed.org/series/CPIAUCSL#0> and USD-Euro exchange rates at [https://www.europarl.europa.eu/RegData/etudes/note/join/2007/379231/IPOL-TRAN_NT\(2007\)379231_EN.pdf](https://www.europarl.europa.eu/RegData/etudes/note/join/2007/379231/IPOL-TRAN_NT(2007)379231_EN.pdf).

flows between Germany and China. We also follow common practice in the trade literature and calculate trade flows between each of eight other high-income countries and China; our “other” high-income countries include the UK, Sweden, Norway, Canada, Australia, New Zealand, Japan, and Singapore.

A.2 Additional Details on Variable Definitions

Retraining Rates. Using the quarterly panel restricted to 2000-2019, we compute the yearly retraining rate as the share of person-years in which a worker participates in subsidized retraining (*erwstat* 10001-10003), both in the aggregate and, as in Figure 1, by age, education, and the nine sectors delineated by Dauth and Eppelsheimer (2020) that correspond to the IAB Establishment Panel’s classification. The lifetime retraining rate cited in the main text is the share of workers in the quarterly panel who complete retraining at some point during 2000-2019.

Retraining Episode Characteristics. We use the retraining episode sample to characterize the distribution of episode length in days in Panel (d) of Figure 1 as well as the share of retraining spells which occur during non-employment cited in the main text, i.e. the share which does not overlap with any employment spell.

Retraining, Switching, and Labor Market Outcomes. We compute all outcomes for the regressions in Tables 2 and 4-5 as well as Supplemental Appendix Tables S2 and S3 in the reduced form sample detailed in Supplemental Appendix A.1.1, building on replication code from Dauth, Findeisen, and Suedekum (2021). *Earnings growth* equals 100 times total 2010 earnings across all jobs, divided by total 2000 earnings. The *share of days employed* is (100 times) the share of calendar days in 2010 during which a worker holds any job, and the *share of days employed in manufacturing* restricts to 3-digit sectors 151-366.

In our labor market effects analysis in Table 4 and Supplemental Appendix Table S3, *sector switching* is defined as a dummy for any employment in 2010 outside of a worker’s initial sector, where we measure sectors using our reduced form sectors in Table 4 and using the 3-digit sectors from the German Classification of Economic Activities, Edition 1993 (WZ 93) in Supplemental Appendix Table S3. In the retraining and switching specifications in Table 2 and Supplemental Appendix Table S2, we instead define sector switching as a dummy for any employment outside of a worker’s initial (reduced form) sector during 2001-2010, so that the outcome is more directly comparable to the retraining dummy, which also covers the same period.

For our linear and logit specifications for $Pr(retrain)$ in Table 5, we define a retraining dummy equal to one if a worker participates in subsidized retraining (*erwstat* 10001-10003) during 2001-2010. *Days retraining* sums the lengths of all such episodes in days over the same period.

Controls. For our retraining, switching, and labor market regressions, we calculate dummies for female, foreign nationality, three education levels – low (no vocational training or university

degree), medium (vocational training), and high (university or university of applied science) – and age bins 24-34, 35-44, 45-54, and 55+. We also calculate controls related to a worker’s primary job in 2000, including dummies for tenure bins (<1, 2-4, 5-9, 10+ years) as defined by [Dauth, Eppelsheimer, and Stüber \(2023\)](#), plant-size bins (1-9, 10-99, 100-499, 500+ workers), four broad manufacturing sector groups, and a dummy for former East German workplaces. We also record total 2000 earnings summed across all jobs as well as, for one specification in Table 2, the share of days employed during 2001-2010.

Post-Non-Employment Sectors. To compute the post-non-employment sector distributions presented in Table 3, we restrict the quarterly panel to 2001-2010 and define each set of consecutive worker-quarters of non-employment as a spell. We then report the share of spells for which the worker’s primary job in the first employed quarter after the non-employment spell is in each of the nine sectors delineated by [Dauth and Eppelsheimer \(2020\)](#), separately for spells during which the worker participates in subsidized retraining and those during which they do not.

Trade Exposure. We calculate the change in net imports, $\Delta \text{net imports}_s$, as the 2001-2010 change in imports minus exports between Germany (or the other high-income countries) and China in reduced form sector s (in Tables 4-5) or 3-digit sector s (in Supplemental Appendix Table S3). We then obtain sectoral import exposure as

$$\text{import exposure}_s \equiv \frac{\Delta \text{net imports}_s}{\text{wage bill}_s},$$

where, following [Dauth, Findeisen, and Suedekum \(2021\)](#), we normalize by the sector’s total year-2000 wage bill in Germany.²⁹

Sectoral and Aggregate Statistics for Calibration. Finally, we calculate sectoral and aggregate statistics for our hat algebra procedure, as described in Section 4.1. We take our quarterly panel, drop any non-employment quarters before workers’ first observed employment and after their last observed employment, group quarters into five-year periods, and then record whether a worker participated in subsidized retraining over the course of a period. We record worker demographics, total daily earnings, employment status, and the model sector of their primary job in the first quarter of the first (base) year of each five-year period.

We then calculate the number of workers and mean total daily earnings by type and model sector in the year 2000, where we define worker types by the intersection of age bins (18-44, 45-65) and education levels (low, medium, and high, defined as above). We separately calculate the number of young labor market entrants, i.e. workers aged 18-44 who first appear in the data as employed during 1996-2000, by skill and model sector in 2000. We finally calculate five-year

²⁹In particular, we use reported daily earnings and episode length to estimate the total annual wage bill by reduced form or 3-digit sector.

Table S2: Retraining and Sector Switching by Worker Type

	(1) Young, Low-Skill	(2) Young, Med.-Skill	(3) Young, High-Skill	(4) Old, Low-Skill	(5) Old, Med.-Skill	(6) Old, High-Skill
Retraining dummy	0.384*** (0.014)	0.375*** (0.006)	0.227*** (0.025)	0.329*** (0.024)	0.362*** (0.011)	0.366*** (0.032)
Demographic Controls	✓	✓	✓	✓	✓	✓
Initial Job Controls	✓	✓	✓	✓	✓	✓
Observations	6,921	44,156	5,796	4,157	19,388	2,681

Notes: Robust standard errors in parentheses; *** indicates $p < 0.01$, ** $p < 0.05$, and * $p < 0.1$. This table presents OLS estimates of a regression of a dummy for switching out of the worker's initial 2000 sector during 2001–2010 on a dummy for participation in subsidized retraining during the same period by worker type, defined by the intersection of two age groups (young: 24–44 in 2000; old: 45+) and three skill levels (low: no vocational training; medium: vocational training; high: university degree). Our sample includes all workers initially employed in manufacturing. Demographic controls include gender, foreign nationality, and an East Germany dummy. Initial, year-2000 job controls include tenure and firm size bins, manufacturing sector group indicators, and log earnings, the latter summed across all jobs.

retraining rates by type and initial employment status, averaging a dummy for participation in subsidized retraining during a five year period over workers and the periods starting in 2000 and 2005 for each type and employment status. We analogously compute five-year retraining rates by model sector over the same periods.

A.3 Retraining and Sector Switching By Worker Type

Table S2 reports the association between retraining and sector switching separately for each of six worker types defined by the intersection of two age groups, young (24–44) and old (45+), and three skill levels: no vocational training, vocational training, and university-educated. The positive correlation between retraining and sector switching documented in Table 2 holds across all six types: retraining raises the probability of a sector switch by 23 to 38 percentage points, depending on the group. Among young workers, the association declines sharply with skill: retraining's effect on sector-switching falls to 23 percentage points for young, high-skill workers.

A.4 Alternative Sector Classification

Tables 4 and 5 in the main text measure import exposure and switching at the level of what we have termed *reduced form sectors*, i.e. 24 slightly-aggregated ISIC Rev. 4 manufacturing sectors. Table S3 re-estimates our key specifications using 3-digit sectors from the German Classification of Economic Activities, Edition 1993 (WZ 93). Point estimates are remarkably stable across the two classifications: the employment, manufacturing employment, earnings growth, and retraining coefficients are virtually identical, and the sector-switching coefficient differs only modestly (3.496

Table S3: Labor Market and Retraining Effects of Chinese Import Exposure: 3-Digit Sectors

	(1)	(2)	(3)	(4)	(5)
<i>IV (2SLS)</i>	Share days employed	Share days in manufacturing	Earnings growth	Sector Switching	Pr(Retrain)
Sectoral import exposure	-2.339*** (0.619)	-4.877*** (1.145)	-2.748** (1.102)	3.496** (1.644)	1.564*** (0.529)
Demographic controls	✓	✓	✓	✓	✓
Initial job controls	✓	✓	✓	✓	✓
Observations	83,099	83,099	83,099	83,099	83,099
Mean of dependent variable	76.1	64.9	80.965	23.6	13.5
First-stage <i>F</i> -statistic	736.18	736.18	736.18	736.18	736.18

Notes: Robust standard errors in parentheses, clustered at the sector level; *** indicates $p < 0.01$, ** $p < 0.05$, and * $p < 0.1$. Columns 1–4 present instrumental variables estimates of (1) for workers employed in manufacturing in 2000 in the SIAB data, where our sectors are the 3-digit German Classification of Economic Activities, Edition 1993 (WZ 93). We measure the share of days employed and share of days employed in manufacturing (unconditional on employment) in 2010, earnings growth as 100 times total 2010 earnings divided by those in 2000, sector switching as a dummy for any employment in 2010 outside of a worker’s initial sector, and retraining as a dummy for participation in subsidized retraining during 2001–2010. Shares and probabilities are multiplied by 100; these outcomes can be interpreted as percentage points.

versus 4.23 in the main specification). Thus, our findings do not depend on the level of sectoral aggregation.

A.5 Decomposition: Trade, Retraining, and Non-Employment

A natural question is whether the positive effect of import exposure on retraining documented in Table 5 operates through a mechanical increase in non-employment—since non-employed workers retrain at far higher rates than employed workers—or whether trade exposure also raises retraining rates, conditional on workers’ employment status. We can decompose the overall effect of import exposure ΔM on retraining probability $\Pr(R)$ in Column (2) of Table 5 as follows,

$$\frac{\partial \Pr(R)}{\partial \Delta M} = \underbrace{(\Pr(R|N) - \Pr(R|E))}_{\text{Via } \Pr(N)} \frac{\partial \Pr(N)}{\partial \Delta M} + \underbrace{\Pr(N)}_{\text{Via } \Pr(R|N)} \frac{\partial \Pr(R|N)}{\partial \Delta M} + \underbrace{(1 - \Pr(N))}_{\text{Via } \Pr(R|E)} \frac{\partial \Pr(R|E)}{\partial \Delta M}, \quad (\text{S.1})$$

where $\Pr(N)$ is the probability of non-employment, $\Pr(R|N)$ and $\Pr(R|E)$ are the probabilities of retraining conditional on non-employment and employment, respectively.

To implement this decomposition in our worker-level sample, we start from the *reduced form sample* discussed in Supplemental Appendix A.1.1 and define a dummy variable N for non-employment such that $N = 1$ if a worker is non-employed for a total timespan greater than T during 2001–2010, where we set T to 1 month in our baseline specification but, given the necessarily arbitrary nature

Table S4: Elements of the Trade-Retraining Decomposition

<i>IV (2SLS)</i>	(1) Pr($R \cap N$)	(2) Pr($R \cap E$)	(3) Pr(N)
Sectoral import exposure	0.02104** (0.00781)	−0.00540*** (0.00192)	0.02539** (0.00934)
Demographic Controls	✓	✓	✓
Initial Job Controls	✓	✓	✓
Observations	83,099	83,099	83,099
Pr(N)		0.460	
Pr($R \cap N$)		0.112	
Pr($R \cap E$)		0.023	

Notes: Robust standard errors in parentheses, clustered at the sector level. *** indicates $p < 0.01$, ** $p < 0.05$, and * $p < 0.1$. Columns 1–3 present instrumental variables estimates of (1) for workers employed in manufacturing in 2000 in the SIAB data and use our reduced form sectors. The outcome in Column 1 is the dummy $R \cap N$, equal to one for workers who were non-employed for a total timespan greater than one month *and* retrained at some point during 2001–2010; Column 2 is the analogous dummy $R \cap E$ for workers who were non-employed for less than one month and retrained; Column 3 is the corresponding non-employment dummy N . Probabilities, unlike in the main text, are reported as decimals. In each case, we instrument for 2001–2010 sectoral import exposure in Germany with an analogous measure calculated using net imports to other high-income countries. Demographic controls include gender, age, education, foreign nationality, and an East Germany dummy. Initial, year-2000 job controls include tenure and firm size bins, manufacturing sector group indicators, and log earnings summed across all jobs.

of such a threshold, also test robustness to 2-week and 3-month thresholds. We then define a dummy variable $R \cap N$ for workers who were non-employed, $N = 1$, by the above definition *and* retrained at some point during 2001–2010, as well as an analogous dummy $R \cap E$ for workers who were employed, $E = 1$, and retrained at some point during this same decade. We calculate $\Pr(N)$, $\Pr(R \cap N)$, and $\Pr(R \cap E)$ as means of these dummy variables. Next, we estimate IV specifications analogous to those in Table 4 but with outcomes N , $R \cap N$, and $R \cap E$ to obtain the respective derivatives with respect to import exposure ΔM . Table S4 displays these estimated inputs to our decomposition.

We then calculate $\Pr(R|N) = \Pr(R \cap N) / \Pr(N) = 0.244$ and $\Pr(R|E) = \Pr(R \cap E) / (1 - \Pr(N)) = 0.042$ as well as, using the chain rule,

$$\begin{aligned} \frac{\partial \Pr(R|N)}{\partial \Delta M} &= \frac{1}{\Pr(N)} \frac{\partial \Pr(R \cap N)}{\partial \Delta M} - \frac{\Pr(R \cap N)}{\Pr(N)^2} \frac{\partial \Pr(N)}{\partial \Delta M} = 0.0323, \\ \frac{\partial \Pr(R|E)}{\partial \Delta M} &= \frac{1}{1 - \Pr(N)} \frac{\partial \Pr(R \cap E)}{\partial \Delta M} + \frac{\Pr(R \cap E)}{(1 - \Pr(N))^2} \frac{\partial \Pr(N)}{\partial \Delta M} = -0.008. \end{aligned}$$

We then plug these derivatives as well as the estimated values of $\Pr(N)$, $\Pr(R|N)$, and $\Pr(R|E)$ from Table S4 into (S.1) to calculate the percentage contribution of each channel to the overall effect of imports on retraining.

The first row of Table S5 displays the results. While the “mechanical” increase in non-employment

Table S5: Decomposition of the Trade-Retraining Effect

Non-employment threshold	Percentage contribution via:		
	Via $\Pr(N)$	Via $\Pr(R N)$	Via $\Pr(R E)$
1 month (baseline)	32.9%	94.9%	−27.8%
2 weeks	32.4%	93.1%	−25.6%
3 months	39.9%	82.7%	−22.5%

Notes: Each cell shows the percentage contribution of the labeled channel to the total effect of import exposure on retraining, $\partial \Pr(R) / \partial \Delta M$, as given by the decomposition in (S.1). Contributions are computed by plugging the derivatives and (conditional) probabilities implied by the 2SLS estimates and sample means from Table S4 into the three terms of the decomposition. Each row sums to 100%. Rows differ in the total number of weeks or months of non-employment over 2001-2010 used as a minimum threshold for defining non-employment at the worker level.

accounts for 32.9% of the overall effect of import exposure on retraining in this baseline decomposition, the increase in retraining conditional on non-employment accounts for 94.9%. Thus, trade clearly appears to alter the incentives for non-employed workers to retrain. The second and third rows display the breakdown based on equivalent specifications but with non-employment thresholds of 2 weeks and 3 months, respectively. The results are broadly similar across these specifications, with the mechanical channel always accounting for between 32 and 40% of the overall effect.

B Model Extensions

B.1 Human Capital

We now extend the simple model of Section 3 to incorporate sector-specific human capital, and allow retraining to work by changing a worker's human capital stock. In Theorem 4 below we show this human capital model is isomorphic to the model of Section 3.

Environment

The environment is identical to Section 3 except that workers now differ in their human capital. Each worker is in one of two states: *experienced*, with human capital $\bar{h}$, or *inexperienced*, with human capital $\underline{h} < \bar{h}$. A worker's state evolves according to her sector choice. An experienced worker who remains in her current sector stays experienced; an experienced worker who switches into a new sector becomes inexperienced. An inexperienced worker who remains in her sector becomes experienced; an inexperienced worker who switches sectors remains inexperienced.

All other elements of the environment carry over from Section 3. In particular, a worker in sector i with human capital h consumes $w_{it}h$, where w_{it} is the real wage per unit of human capital. With log utility, this yields a flow payoff of $\log w_{it} + \log h$. We retain the non-pecuniary switching cost C and retraining-switching complementarity Γ from the main text, so that switching costs

and the gains from retraining now have both a pecuniary (human capital) and a non-pecuniary component.

Retraining

As in the main text, the worker simultaneously decides whether to switch sectors and whether to retrain. Retraining carries an idiosyncratic utility cost $\sigma\eta - F$, where η is drawn from a logistic distribution and F is a deterministic cost. The deterministic benefit of retraining operates through human capital. Let $f(h)$ be an increasing function that captures the effect of retraining on human capital. A worker who retraines and whose destination state has human capital h instead arrives with human capital $f(h)$ in the following period; this effect then dissipates, so that in subsequent periods the worker again has human capital determined by her experience state. The result is that the one-period benefit of retraining for a worker arriving in state $h \in \{\underline{h}, \bar{h}\}$ is $\beta(\log f(h) - \log h)$.

Bellman equations

A worker's state is a pair (i, s) where i is her sector and $s \in \{e, u\}$ denotes experienced or inexperienced. Consider first the case without retraining. The value of an experienced worker in sector i satisfies

$$v_{it}^e = \log(w_{it}\bar{h}) + \mathbb{E} \left[\max_j \{ \beta v_{it+1}^e \mathbb{1}\{j = i\} + \beta v_{jt+1}^u \mathbb{1}\{j \neq i\} + v\epsilon_{ij} - C \mathbb{1}\{j \neq i\} \} \right], \quad (\text{S.2})$$

while the value of an inexperienced worker satisfies

$$v_{it}^u = \log(w_{it}\underline{h}) + \mathbb{E} \left[\max_j \{ \beta v_{it+1}^e \mathbb{1}\{j = i\} + \beta v_{jt+1}^u \mathbb{1}\{j \neq i\} + v\epsilon_{ij} - C \mathbb{1}\{j \neq i\} \} \right]. \quad (\text{S.3})$$

Notice that the continuation values inside the expectation are identical across the two types. Inspection shows that $v_{it}^e - v_{it}^u = \log(\bar{h}/\underline{h})$ for all i and t : the two types differ only in their flow payoffs.

Adding retraining, the full Bellman equation for a worker in state s becomes

$$v_{it}^s = \log(w_{it}h_s) + \mathbb{E} \left[\max_j \{ \beta \hat{v}_{jt+1}(i) + v\epsilon_{ij} - C \mathbb{1}\{j \neq i\} \right. \\ \left. + \max\{\Gamma \mathbb{1}\{j \neq i\} + \beta \Delta_f(j, i) + \sigma\eta - F, 0\} \} \right], \quad (\text{S.4})$$

where $h_s \in \{\underline{h}, \bar{h}\}$ is the human capital associated with state s , the continuation value

$$\hat{v}_{jt+1}(i) \equiv v_{it+1}^e \mathbb{1}\{j = i\} + v_{jt+1}^u \mathbb{1}\{j \neq i\}$$

captures the experience transition, and the one-period retraining benefit in destination state h is

$$\Delta_f(j, i) \equiv \begin{cases} \log f(\bar{h}) - \log \bar{h} & \text{if } j = i, \\ \log f(\underline{h}) - \log \underline{h} & \text{if } j \neq i. \end{cases}$$

Since the continuation values are the same for both types, the sector and retraining choices are also the same: the experience state affects levels but not decisions.

Isomorphism

Theorem 4. *The human capital model described above is isomorphic to the baseline model in Section 3. In particular, define composite switching costs, complementarities, and retraining costs as*

$$\tilde{C} \equiv C + \beta \log(\bar{h}/\underline{h}), \quad (\text{S.5})$$

$$\tilde{\Gamma} \equiv \Gamma + \beta [\log(\bar{h}/\underline{h}) - \log(f(\bar{h})/f(\underline{h}))], \quad (\text{S.6})$$

$$\tilde{F} \equiv F - \beta \log(f(\bar{h})/\bar{h}). \quad (\text{S.7})$$

Then the sector choice probabilities μ_{ij} , retraining rates ρ_i , and the evolution of employment l_{it} in the human capital model are identical to those in the baseline model with switching cost $\tilde{C}$, complementarity $\tilde{\Gamma}$, and retraining cost $\tilde{F}$. Theorems 1–3 therefore apply to the human capital model with these parameter values.

Proof. Substitute $v_{it+1}^e = v_{it+1}^u + \log(\bar{h}/\underline{h})$ into the continuation value of (S.4) to obtain

$$\hat{v}_{jt+1}(i) = v_{jt+1}^u + \log(\bar{h}/\underline{h}) \cdot \mathbb{1}\{j = i\},$$

and therefore

$$\beta \hat{v}_{jt+1}(i) - C \mathbb{1}\{j \neq i\} = \beta v_{jt+1}^u + \beta \log(\bar{h}/\underline{h}) - \tilde{C} \mathbb{1}\{j \neq i\},$$

where $\tilde{C} = C + \beta \log(\bar{h}/\underline{h})$. Next, decompose the retraining benefit as

$$\beta \Delta_f(j, i) = \beta (\log f(\bar{h}) - \log \bar{h}) + \beta [\log f(\underline{h}) - \log \underline{h} - \log f(\bar{h}) + \log \bar{h}] \mathbb{1}\{j \neq i\}.$$

The term in brackets equals $\tilde{\Gamma} - \Gamma$ after dividing by β . Defining $\tilde{F} = F - \beta (\log f(\bar{h}) - \log \bar{h})$, the full retraining payoff becomes

$$\max\{\Gamma \mathbb{1}\{j \neq i\} + \beta \Delta_f(j, i) + \sigma\eta - F, 0\} = \max\{\tilde{\Gamma} \mathbb{1}\{j \neq i\} + \sigma\eta - \tilde{F}, 0\}.$$

Substituting both expressions into (S.4) and collecting constants yields

$$\begin{aligned} v_{it}^s = \log(w_{it}h_s) + \beta \log(\bar{h}/\underline{h}) + \mathbb{E} \Big[& \max_j \{ \beta v_{jt+1}^u + v\epsilon_{ij} - \tilde{C} \mathbb{1}\{j \neq i\} \\ & + \max\{\tilde{\Gamma} \mathbb{1}\{j \neq i\} + \sigma\eta - \tilde{F}, 0\} \} \Big]. \end{aligned}$$

The term $\log(w_{it}h_s) + \beta \log(\bar{h}/\underline{h})$ is a type-dependent constant that does not affect comparisons across destinations j . The remaining expression has the same structure as the baseline Bellman equation (4), with v_{jt+1}^u playing the role of v_{jt+1} and the parameters $(\tilde{C}, \tilde{\Gamma}, \tilde{F})$ replacing (C, Γ, F) . Since the choice probabilities are determined entirely by within-destination comparisons, μ_{ij} and ρ_i are the same as in the baseline model. $\square$

The composite parameters $\tilde{C}$, $\tilde{\Gamma}$, and $\tilde{F}$ each have two components: the non-pecuniary costs C , Γ , and F from the main text, and the human capital terms. The human capital contribution to $\tilde{\Gamma}$ is positive whenever f is concave, so that retraining compresses the gap between experienced and inexperienced workers. The baseline model is the special case $\bar{h} = \underline{h}$; a pure human capital model sets $C = \Gamma = 0$.

Theorem 4 tells us we can remain agnostic about the exact mechanism by which retraining works: via human capital, non-pecuniary switching costs, or some combination of the two. The key assumption in generating this result is that the effects of retraining are transitory and dissipate within a period. Without this, retraining would have long-run effects on human capital and we would have to keep track of a richer state space. But, given that Section 1 documented that most retraining courses are relatively short and that our model periods last five years, the assumption that retraining only has transitory effects does not seem unreasonable.

C Additional Calibration Details and Counterfactual Results

C.1 Switching costs

We now describe how we parameterize and recover sector switching costs. Since the procedure works entirely within a worker type, we drop the ka superscript throughout. We set $C_{ii} = 0$ and $C_{i0} = 0$ for all i , so that remaining in a sector and entering non-employment are costless. Letting $M(i) \in \{0, 1\}$ indicate manufacturing, the cost of moving from i to j is

$$C_{ij} = C_i^o + C_j^d + C_{MN}M(i)(1 - M(j)) + C_{NM}(1 - M(i))M(j) + C_\delta\delta(i, j), \quad (\text{S.8})$$

where $\delta(i, j)$ is a measure of the skill distance between sectors i and j . For workers who are initially non-employed, we leave C_{0i} free.

We measure skill distance using the occupational task content weights developed by [Dengler, Matthes, and Paulus \(2014\)](#) based on a German expert database. In particular, we average their 3-digit-occupation-level weights for the year 2011 – the closest year to our sample period available – on analytical non-routine tasks, interactive non-routine tasks, cognitive routine tasks, manual routine tasks and manual non-routine tasks at the model sector level using employment shares in the first quarter of 2000 as weights. We then take the log Euclidean distance between the five average task weights of each pair of model sectors to obtain $\delta(i, j)$.³⁰

³⁰Note that we set skill distance equal to zero for same-sector transitions as well as those in and out of non-employment.

Our goal is to infer the cost parameters in (S.8) from the observed transition matrix μ_{ij} . To motivate our approach, first consider a model without retraining. Then the (log) probability a worker switches from sector i to sector $j \neq i$ is:

$$\log \mu_{ij} = \left(\frac{1}{\nu}\right) (\beta v_j - U_i) - \left(\frac{1}{\nu}\right) \left(C_i^o + C_j^d + C_{MN}M(i)(1 - M(j)) + C_{NM}(1 - M(i))M(j) + C_\delta \delta(i, j) \right), \quad (\text{S.9})$$

while the probabilities of remaining in i or entering non-employment are:

$$\log \mu_{ii} = \left(\frac{1}{\nu}\right) (\beta v_i - U_i), \quad \log \mu_{i0} = \left(\frac{1}{\nu}\right) (\beta v_0 - U_i). \quad (\text{S.10})$$

With the normalization $v_0 = 0$, (S.10) identifies both v_i and U_i for all i , which pins down the non-cost terms in (S.9) and allows us to estimate the switching cost parameters — up to the proportional constant ν — via PPML with origin and destination fixed effects, origin fixed effects interacted with a dummy for staying in the same sector, bilateral dummies for transitions between non-manufacturing and manufacturing, the reverse, and for workers staying within non-manufacturing and manufacturing, and skill distance. The fitted transition matrix matches the data closely (correlation > 0.99) and in particular exactly matches the matrix diagonal and flows into non-employment by construction.

The presence of retraining means that choice probabilities do not take exactly the form in (S.9). However, retraining does not affect the relative attractiveness of remaining in sector i versus entering non-employment, so

$$\log \mu_{ii} - \log \mu_{i0} = \left(\frac{\beta}{\nu}\right) (v_i - v_0),$$

which, with the normalization $v_0 = 0$, allows us to recover all the values v_i . Given these values, we adopt an indirect-inference approach to recover switching costs. We run the aforementioned PPML regression using transition probabilities measured between 1995-2000, and record the resulting coefficients.³¹ Then, given a candidate Θ and the values v_i , we solve for the switching cost parameters such that the same regression run on model-implied choice probabilities yields the same coefficients. We repeat this for every new value of Θ , although in practice only ν matters much for the inferred costs.

³¹We calculate bilateral transition probabilities between the *model sectors* in Supplemental Appendix Table S1 from our quarterly worker panel, described in Supplemental Appendix A.1.1, where we keep workers between their first and last quarters of employment and then calculate bilateral transition probabilities between the slightly-aggregated ISIC Rev. 4 sectors used throughout based on workers' sectors in the first quarters of 1995 and 2000. Manufacturing includes our numbered sectors 6-24, while non-manufacturing is sectors 2-5 and 25.

C.2 Decomposition of retraining gains

Consider an employed worker of type ka in sector i at time t who has drawn retraining shock η and sector preference shocks ϵ , and who chooses to retrain. Define $\mathcal{G}_{it}^{ka}$ as the difference between this worker's expected lifetime utility when retraining is available versus when it is not.

Without retraining, the worker chooses sector h^* to solve $\max_j \{\beta v_{jt+1}^{ka} + v\epsilon_{ij} - C_{ij}^{ka}\}$. Let $S_0 = \mathbb{1}\{h^* \neq i, 0\}$. With retraining, switching to a new productive sector $j \neq i, 0$ yields an additional Γ^{ka} , so the worker chooses j^* to solve $\max_j \{\beta v_{jt+1}^{ka} + v\epsilon_{ij} - C_{ij}^{ka} + \Gamma^{ka} \mathbb{1}\{j \neq i\} \mathbb{1}\{j \neq 0\}\}$. Let $S_1 = \mathbb{1}\{j^* \neq i, 0\}$. When $\Gamma^{ka} \geq 0$, retraining weakly increases the incentive to switch, so $S_1 \geq S_0$.

Write $V_j = \beta v_{jt+1}^{ka} + v\epsilon_{ij} - C_{ij}^{ka}$ for the common payoff component. The gain from retraining is $\mathcal{G}_{it}^{ka} = (\sigma\eta - F^{ka}) + (V_{j^*} + S_1\Gamma^{ka}) - V_{h^*}$. Three cases arise. If neither scenario involves switching ($S_0 = S_1 = 0$), then $j^* = h^* = i$ or $j^* = h^* = 0$, and the gain is simply $\sigma\eta - F^{ka}$. If both do ($S_0 = S_1 = 1$), then j^* maximizes $V_j + \Gamma^{ka}$ while h^* maximizes V_j ; but since Γ^{ka} is common across all $j \neq i, 0$, the ranking is unchanged and $j^* = h^*$, giving a gain of $\sigma\eta - F^{ka} + \Gamma^{ka}$. Finally, if retraining induces switching ($S_0 = 0, S_1 = 1$), then the gain is $\sigma\eta - F^{ka} + (V_{j^*} - V_{h^*}) + \Gamma^{ka}$. Combining yields:

$$\begin{aligned} \mathcal{G}_{it}^{ka} = & \underbrace{(\sigma\eta - F^{ka})}_{\text{Direct}} + \underbrace{S_0 S_1 \Gamma^{ka}}_{\text{Switching: intensive}} \\ & + \underbrace{(1 - S_0) S_1 \left(\beta(v_{j^*t+1} - v_{h^*t+1}) + v(\epsilon_{ij^*} - \epsilon_{ih^*}) - (C_{ij^*}^{ka} - C_{ih^*}^{ka} - \Gamma^{ka}) \right)}_{\text{Switching: extensive}}, \end{aligned}$$

as claimed.

C.3 Full calibration results

Table S6 shows the calibration targets (with standard errors) and corresponding model moments. The calibrated model perfectly fits all the moments apart from those related to the China shock in Panel A.

C.4 The calibrated China shock

The China shock enters the model as a vector of proportional changes in Chinese productivity across manufacturing sectors. As described in Section 4, we choose this vector so that the model-implied changes in German net imports from China match those predicted by a version of our first-stage regressions without controls. The resulting shocks are large and heterogeneous: over ten years, the median sector experiences an annualized productivity increase of roughly 1.6 log points, but values range from -1.3 to 3.9 across sectors.

These exogenous productivity changes do not, however, correspond directly to measured total factor productivity (TFP). As in [Caliendo, Dvorkin, and Parro \(2019\)](#), the measured log TFP change in sector i is

$$\Delta \log \text{TFP}_i = \gamma_{ni} \cdot CS_i - \left(\frac{1}{\sigma_i} \right) \log \hat{\pi}_{nms}, \quad (\text{S.11})$$

where CS_s is the exogenous China shock, γ_{ni} is the value-added share, σ_i is the trade elasticity, and the country index n refers to China. Figure S.1 shows the annualized growth rate of measured TFP implied by this expression, computed over the first two model periods (10 years). Measured TFP growth is positive in all but two sectors and averages just over 4% per year. This is somewhat lower than other estimates of Chinese manufacturing TFP growth (Brandt, Van Biesebroeck, and Zhang 2012), but note that in our calibration productivity is held constant in all other countries: so the growth rates in Figure S.1 should be understood as excess TFP growth relative to a global average.

C.5 Comparison to ν estimates in the literature

Our calibration yields $\nu = 0.307$. While a number of papers estimate similar parameters, it is not obvious how to compare them given differences in time horizon (quarterly vs. annual vs five years in our case) across papers. To make progress, we focus on the simple model, and starting from a

Table S6: Calibration Results

	Data	SE	Model
<i>Panel A: China shock effects</i>			
Earnings	-2.748	(0.929)	-4.182
Employment	-2.340	(0.701)	-2.116
Manufacturing employment	-4.880	(1.610)	-6.088
Sector switching	4.230	(2.380)	6.625
Retraining	1.564	(0.697)	1.549
<i>Panel B: Retraining rates</i>			
<i>Young</i>			
Low, Non-employed	0.310	(0.002)	0.310
Low, Employed	0.206	(0.001)	0.206
Medium, Non-employed	0.262	(0.001)	0.262
Medium, Employed	0.085	(0.000)	0.085
High, Non-employed	0.162	(0.003)	0.162
High, Employed	0.045	(0.001)	0.045
<i>Old</i>			
Low, Non-employed	0.228	(0.007)	0.228
Low, Employed	0.038	(0.001)	0.038
Medium, Non-employed	0.278	(0.002)	0.278
Medium, Employed	0.047	(0.000)	0.047
High, Non-employed	0.256	(0.006)	0.256
High, Employed	0.034	(0.001)	0.034
<i>Panel C: Retraining-switching coefficients</i>			
<i>Young</i>			
Low	0.384	(0.016)	0.384
Medium	0.375	(0.019)	0.375
High	0.227	(0.038)	0.227
<i>Old</i>			
Low	0.329	(0.035)	0.329
Medium	0.362	(0.013)	0.362
High	0.366	(0.044)	0.366

symmetric baseline, consider the effects of a permanent shock to real wages in sector j given by $d \log w_j$. The resulting change in switching probabilities is:

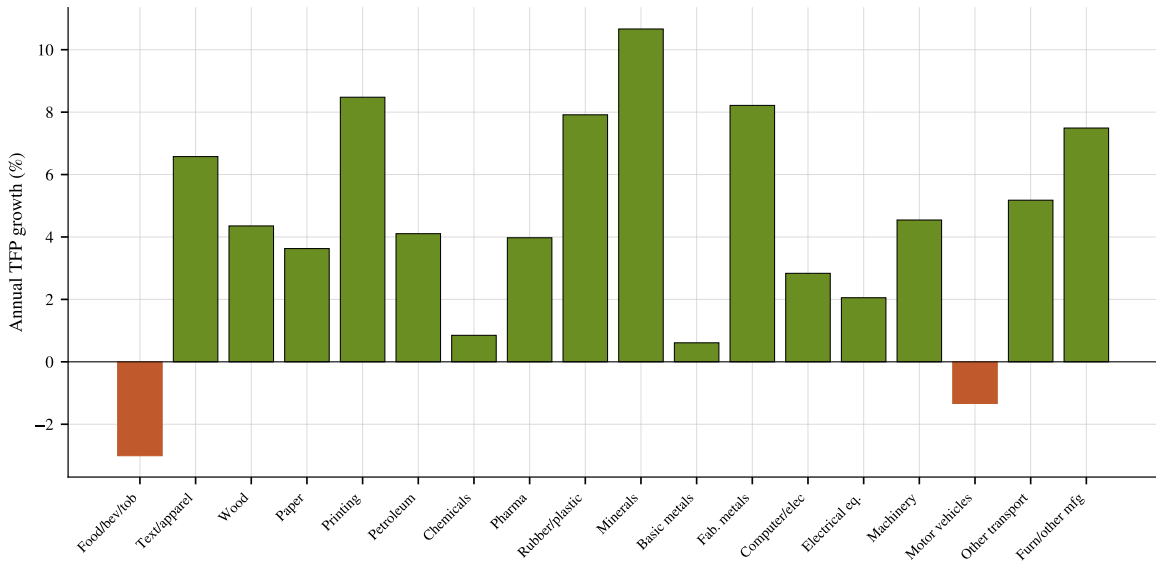
$$d \log \left(\frac{\mu_{ij}}{\mu_{ii}} \right) = \left(\frac{\tilde{\rho}}{\tilde{\rho} + \tilde{m}} \right) \left(\frac{1}{\nu \tilde{\rho}} \right) d \log w_j, \quad (\text{S.12})$$

where $\tilde{\rho} = \beta^{-1}(1 - \beta)$ is the discount rate and $\tilde{m}$ is the switching rate defined in Section 3. We refer to the elasticity of $\frac{\mu_{ij}}{\mu_{ii}}$ with respect to real wages as the implied switching sensitivity. Focusing on this quantity is helpful because we expect both the switching rate $\tilde{m}$ and the discount rate $\tilde{\rho}$ to scale roughly linearly with the length of one model period, implying that the first term in parentheses does not vary across time horizons. Then we can compare this implied switching sensitivity using only each paper's values for ν and $\tilde{\rho}$.

Table S7 shows this comparison, plugging in $\frac{\tilde{\rho}}{\tilde{\rho} + \tilde{m}} = 0.373$ based on the value for $\tilde{\rho}$ implied by our calibration (0.185) and the value for $\tilde{m}$ we find in the German data in Column (2) of Table 8 (0.311). With the caveat that other papers incorporate other aspects of the decision to switch sectors — human capital specificity, occupation switching, geography — our estimate of ν implies a degree of mobility that is in line with the literature.

C.6 Additional Counterfactual Results

Figure S.1: Chinese Measured TFP Growth Implied by the Model



Notes: Bars display the annualized growth rate of measured sectoral TFP from (S.11) implied by the China shock, computed over the first two model periods (10 years).

Table S7: Switching Sensitivity Implied by Estimates of ν

Paper	ν	$\tilde{\rho}$	Implied switching sensitivity
Artuç, Chaudhuri, and McLaren (2010)	2.897	0.030	4.292
Dix-Carneiro (2014)	2.149	0.050	3.472
Traiberman (2019)	0.700	0.040	13.321
Caliendo, Dvorkin, and Parro (2019)	5.340	0.010	6.985
This paper	0.307	0.185	6.567

Note: Artuç, Chaudhuri, and McLaren (2010) estimate from Table 3 Panel II; Dix-Carneiro (2014) from Table A.V; Traiberman (2019) from Column (1) of Table 5; Caliendo, Dvorkin, and Parro (2019) from “Elasticities”, pp764.

Table S8: Effects of the China Shock Across Retraining Scenarios (Partial Equilibrium)

	No retraining	Baseline	2x	3x
Retraining rate	0.0%	11.5%	22.9%	34.4%
Switching rate	28.2%	31.1%	33.8%	36.2%
<i>Welfare: aggregate</i>				
Mean EV	0.485	0.448	0.423	0.405
Mean EV (mfg)	0.279	0.263	0.249	0.239
<i>Welfare: distribution</i>				
Std(EV)	1.369	1.236	1.135	1.059
Std(EV) (mfg)	2.919	2.634	2.419	2.255
% of mfg. workers with EV < 0	37.0	37.0	37.0	37.0
Min EV	-13.702	-12.131	-10.972	-10.358
<i>Aggregate employment dynamics</i>				
Aggregate employment (pp change), year 5	0.063	0.065	0.064	0.061
Aggregate employment (pp change), LR	0.120	0.116	0.102	0.087
Mfg. employment (pp change), year 5	-0.178	-0.189	-0.201	-0.215
Mfg. employment (pp change), LR	-0.191	-0.219	-0.248	-0.275
<i>Sectoral employment dynamics (mfg)</i>				
Short-run reallocation	4.855	4.900	5.011	5.132
Long-run reallocation	7.620	7.100	6.704	6.412
% of LR. at year 5	63.7	68.9	74.7	80.0

Note: This table repeats the experiments of Table 8 in partial equilibrium, holding wages fixed at the path from the baseline retraining scenario. The effects of the China shock are always defined as differences between the economy with and without this shock, so that direct effects of changes in retraining costs are differenced out. Means and standard deviations are weighted by baseline employment. “Mfg” refers to manufacturing. Our measure of reallocation at year t is the average of $|(l_{mt} - l_{m0})|$, where m indexes manufacturing sectors and employment l_{mt} is measured as a percentage of total manufacturing employment in period t . Short run refers to $t = 1$ and long run to $t \rightarrow \infty$.

Table S9: Heterogeneous Effects of the China Shock by Worker Group

	No retraining	Baseline	2x	3x
<i>Panel A: Young</i>				
Retraining rate	0.0%	13.5%	25.6%	36.8%
Switching rate	32.5%	35.9%	38.7%	41.0%
Mean EV	0.617	0.591	0.574	0.563
Mean EV (mfg)	0.457	0.464	0.466	0.468
Std(EV)	1.452	1.302	1.180	1.086
Std(EV) (mfg)	3.250	2.914	2.642	2.430
% of mfg. workers with EV < 0	36.7	36.0	36.0	36.0
Min EV	-13.599	-12.131	-10.883	-10.122
<i>Panel B: Old</i>				
Retraining rate	0.0%	6.6%	16.7%	28.8%
Switching rate	18.3%	20.0%	22.3%	24.6%
Mean EV	0.114	0.114	0.115	0.115
Mean EV (mfg)	-0.120	-0.109	-0.092	-0.074
Std(EV)	1.023	0.987	0.930	0.870
Std(EV) (mfg)	2.032	1.962	1.848	1.729
% of mfg. workers with EV < 0	38.8	38.8	38.8	38.8
Min EV	-8.983	-8.734	-8.299	-7.821
<i>Panel C: Low skill</i>				
Retraining rate	0.0%	18.2%	29.5%	39.3%
Switching rate	33.1%	37.3%	39.7%	41.6%
Mean EV	0.295	0.277	0.274	0.274
Mean EV (mfg)	-0.041	-0.030	-0.013	0.004
Std(EV)	1.482	1.318	1.220	1.142
Std(EV) (mfg)	2.929	2.604	2.410	2.256
% of mfg. workers with EV < 0	42.1	40.4	40.4	40.4
Min EV	-12.287	-10.309	-9.325	-8.642
<i>Panel D: Medium skill</i>				
Retraining rate	0.0%	11.0%	22.7%	34.4%
Switching rate	27.4%	30.3%	33.1%	35.6%
Mean EV	0.497	0.476	0.461	0.451
Mean EV (mfg)	0.333	0.338	0.341	0.343
Std(EV)	1.361	1.238	1.126	1.037
Std(EV) (mfg)	2.928	2.663	2.420	2.225
% of mfg. workers with EV < 0	36.7	36.7	36.7	36.7
Min EV	-13.599	-12.131	-10.883	-9.921
<i>Panel E: High skill</i>				
Retraining rate	0.0%	6.0%	15.9%	28.1%
Switching rate	27.0%	28.3%	30.1%	32.1%
Mean EV	0.492	0.493	0.490	0.485
Mean EV (mfg)	0.213	0.241	0.265	0.285
Std(EV)	1.136	1.088	1.020	0.954
Std(EV) (mfg)	2.553	2.444	2.288	2.135
% of mfg. workers with EV < 0	35.0	33.1	33.1	33.1
Min EV	-12.250	-11.663	-10.874	-10.122

Note: Each column shows effects of the China shock under different retraining cost scenarios, by worker group. The effects of the China shock are always defined as differences between the economy with and without this shock, so that direct effects of changes in retraining costs are differenced out. Means and standard deviations are weighted by baseline employment within each group. “Mfg” refers to manufacturing.

Table S10: Effects of the US Tariff Shock Across Retraining Scenarios

	No retraining	Baseline	2x	3x
Retraining rate	0.0%	11.4%	25.8%	33.1%
Switching rate	28.2%	31.2%	34.5%	35.9%
<i>Welfare: aggregate</i>				
Mean EV	-0.87	-0.85	-0.85	-0.85
Mean EV (mfg)	-0.95	-0.93	-0.92	-0.92
<i>Welfare: distribution</i>				
Std(EV)	0.48	0.46	0.45	0.44
Std(EV) (mfg)	0.70	0.66	0.63	0.61
% of mfg. workers with EV < 0	98.7	98.7	99.4	99.4
Min EV	-2.69	-2.57	-2.42	-2.38
<i>Aggregate employment dynamics</i>				
Aggregate employment (pp change), year 5	-0.13	-0.15	-0.15	-0.15
Aggregate employment (pp change), LR	-0.20	-0.22	-0.22	-0.21
Mfg. employment (pp change), year 5	-0.05	-0.05	-0.05	-0.05
Mfg. employment (pp change), LR	-0.09	-0.09	-0.09	-0.09
<i>Sectoral employment dynamics (mfg)</i>				
Short-run reallocation	0.76	0.78	0.79	0.80
Long-run reallocation	1.20	1.19	1.17	1.16
% of LR. at year 5	63.2	65.0	67.5	68.7

Note: Each column shows effects of the US tariff shock — a 20% increase in Germany’s iceberg cost of selling to the US — under different retraining cost scenarios. The effects of the tariff shock are always defined as differences between the economy with and without the tariff, so that direct effects of changes in retraining costs are differenced out. Means and standard deviations are weighted by baseline employment. “Mfg” refers to manufacturing. Our measure of reallocation at year t is the average of $|(l_{mt} - l_{m0})|$, where m indexes manufacturing sectors and employment l_{mt} is measured as a percentage of total manufacturing employment in period t . Short run refers to $t = 1$ and long run to $t \rightarrow \infty$.